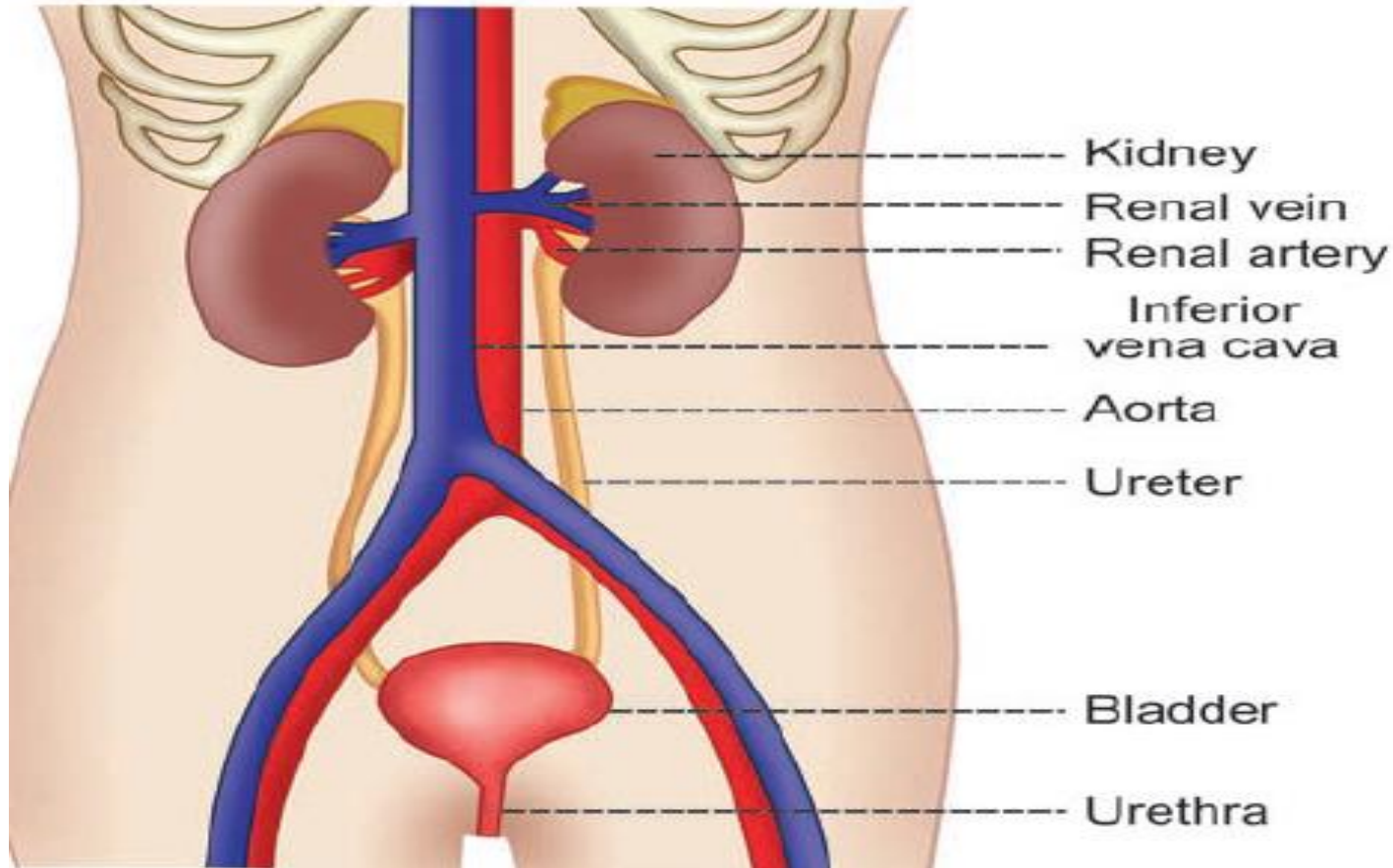


Renal Physiology



By: B.G.

Contents

- 1. Functional structure of the kidney***
- 2. The Nephron***
- 3. The 3-processes of Urine formation***
 - I. Glomerular filtration,***
 - II. Renal reabsorption,***
 - III. Renal secretion***
- 4. Control of urine volume by concentration & dilution processes***
- 5. Acid – base balance***

Functional structures of the kidney

A. Shape: **Bean shaped**

Wt: both weigh ~ 300g

Blood supply: ~ 26%
CO

B. three layers

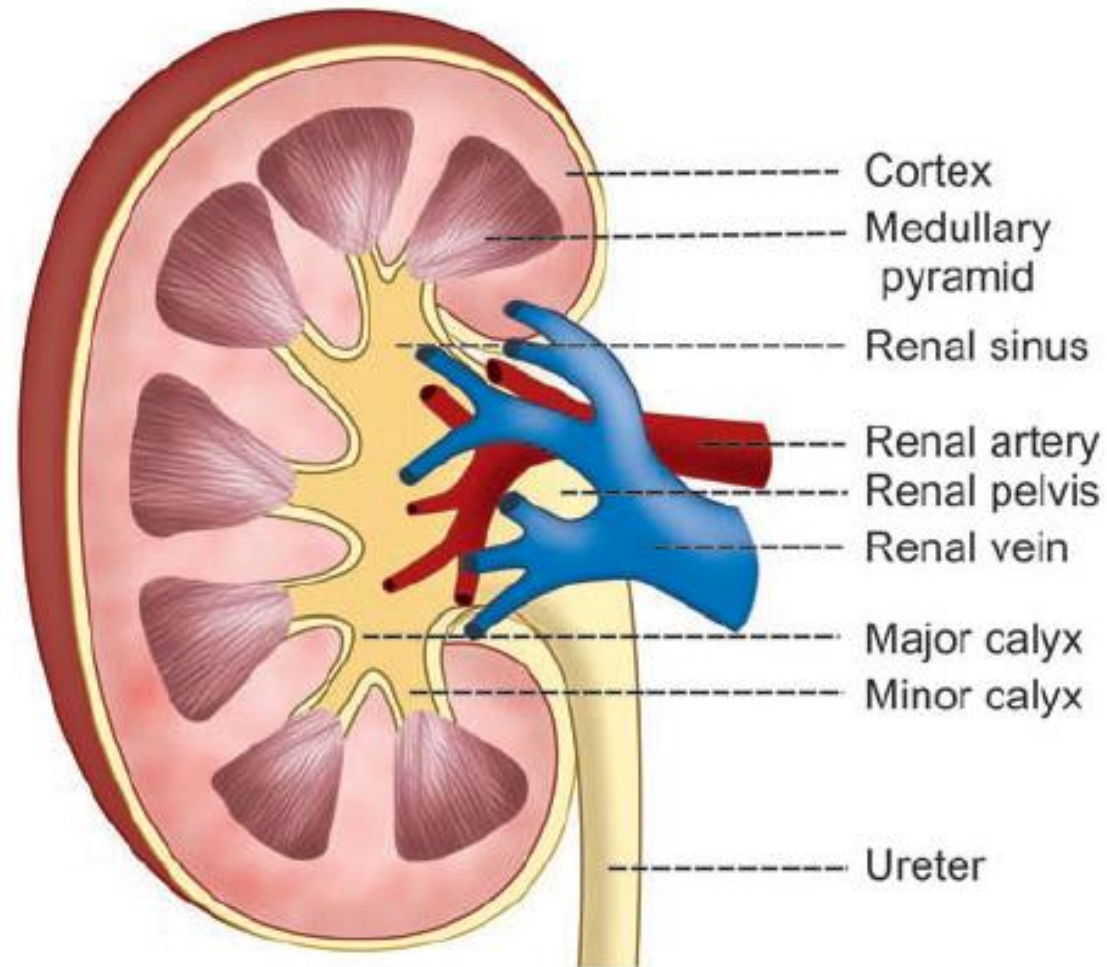
Cortex: light outer

Medulla: dark inner

Renal sinus

C. Deeper structures

**Nerves, blood vessels
and lymphatic's** enter
through *hilum*



Gross anatomy of the kidney.

Function of Kidney

A. Homeostasis (regulation)

- H₂O and electrolyte balance
- acid-base balance in the body
- Regulation of BP
 - ✓ By regulating the volume of extracellular fluid
 - ✓ Through **renin-angiotensin mechanism**

B. Excretion

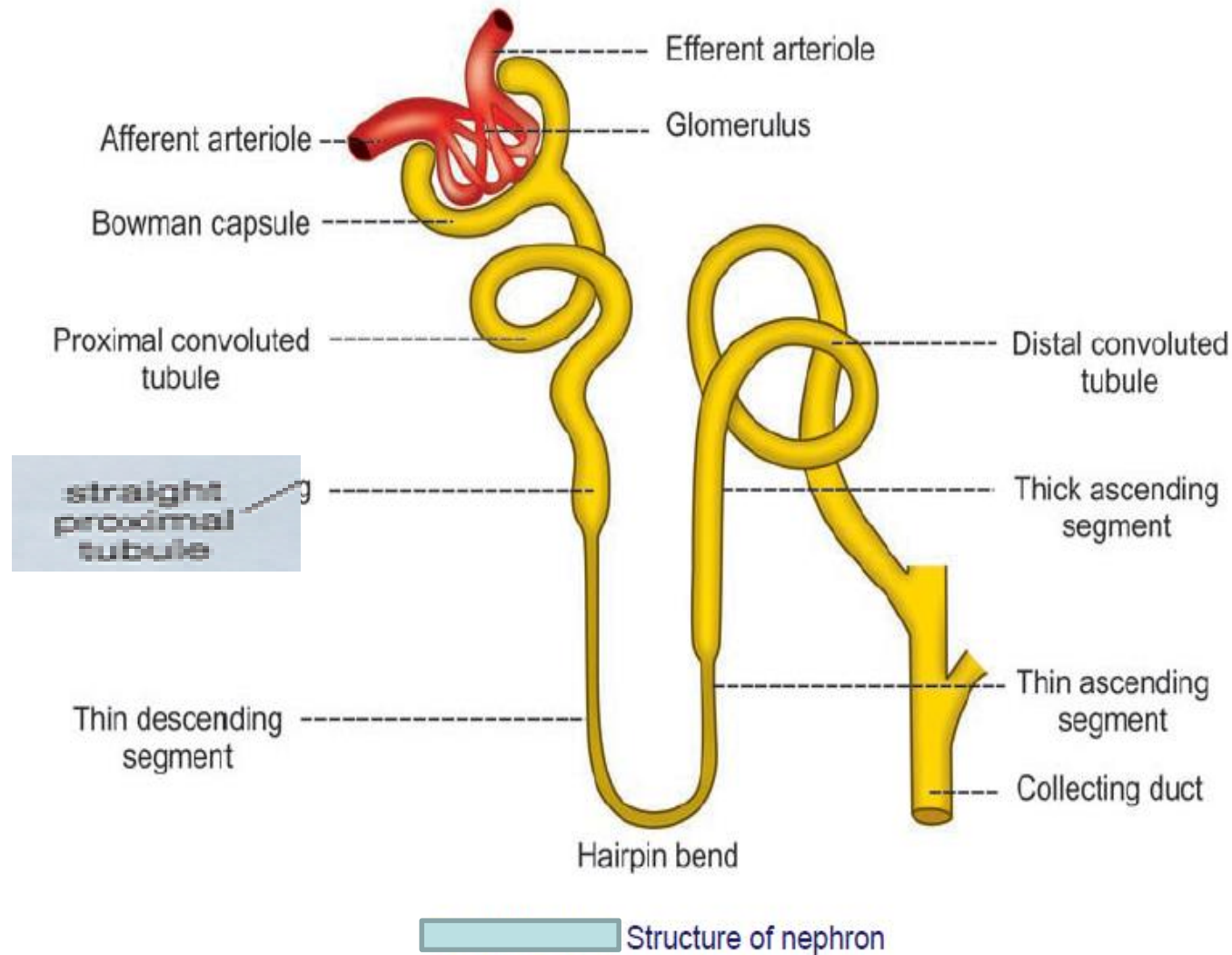
- Excretion of metabolic wastes in urine
(**creatinine, urea, uric acid, drugs, hormones** etc)

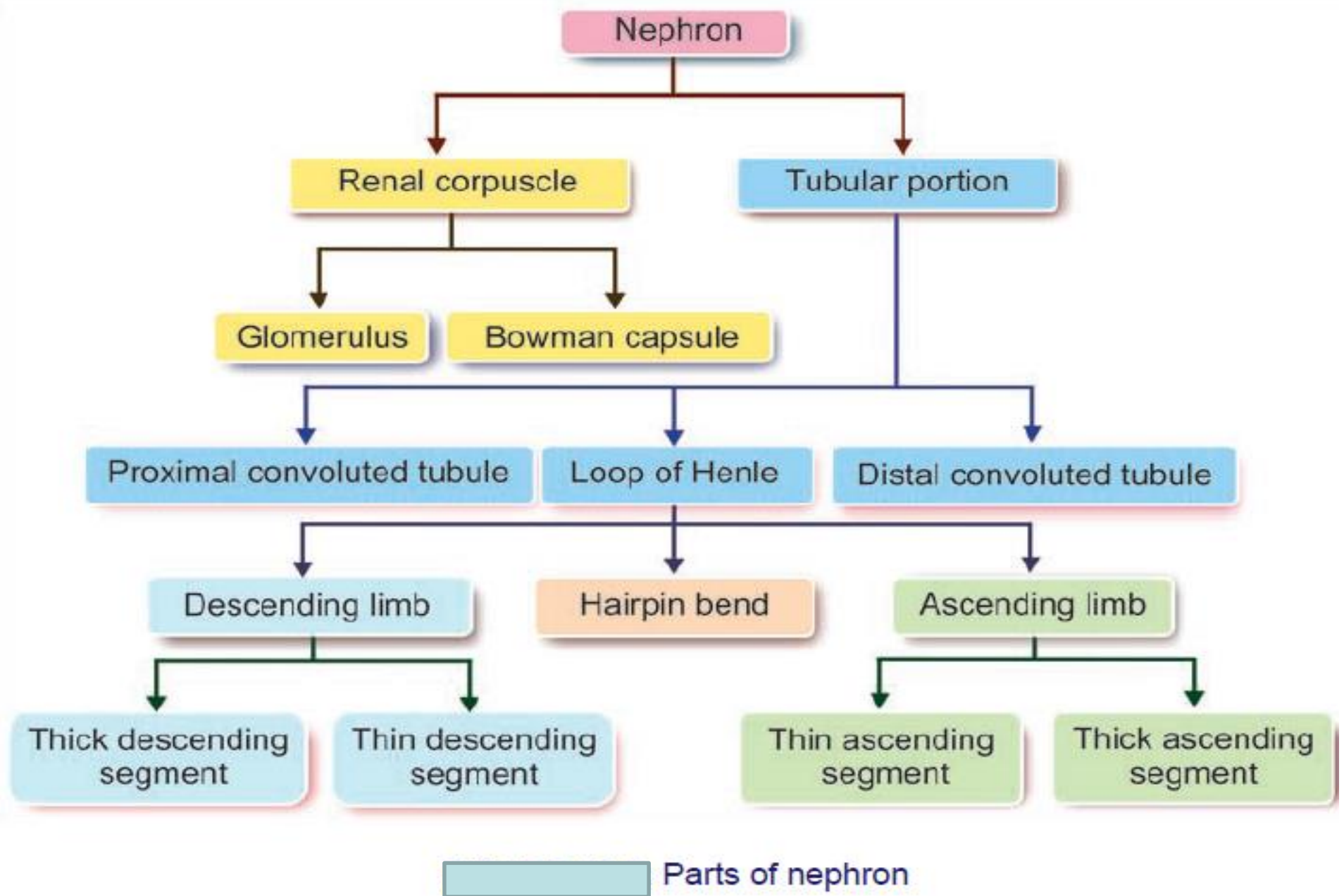
C. Endocrine function

- Renin, erythropoietin, Vit. D₃ production

Nephron

- **Nephron** is defined as the structural and functional unit of kidney.
- Each kidney consists of 1 to 1.3 millions of nephrons.
- Their numbers decrease in old ages (>40 years). But, the remaining can function properly





Different parts of the Nephron

A. Nephron consists of **Renal corpuscle**(Glomerulus + Bowman's capsule) and **renal tubule**.

Glomerulus: is a capillary tuft and is a place of filtration.

B. Proximal convoluted tubule: place of major absorption

C. Loop of Henle:

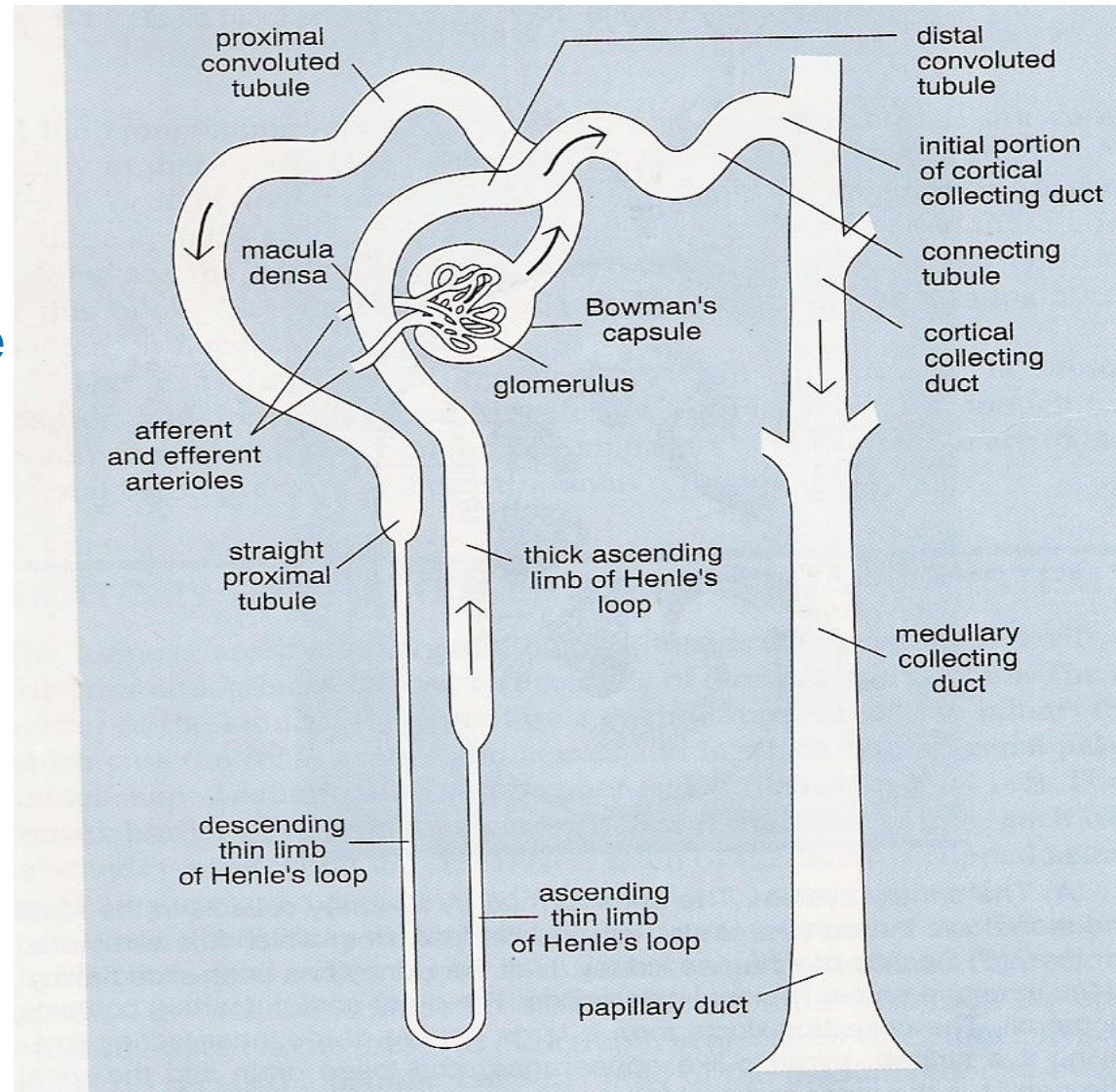
- Descending thin segment
 - **hyperosmolar**
- Ascending thick segment
 - **hyposmolar**

D. Distal convoluted tubule:

- diluting segment

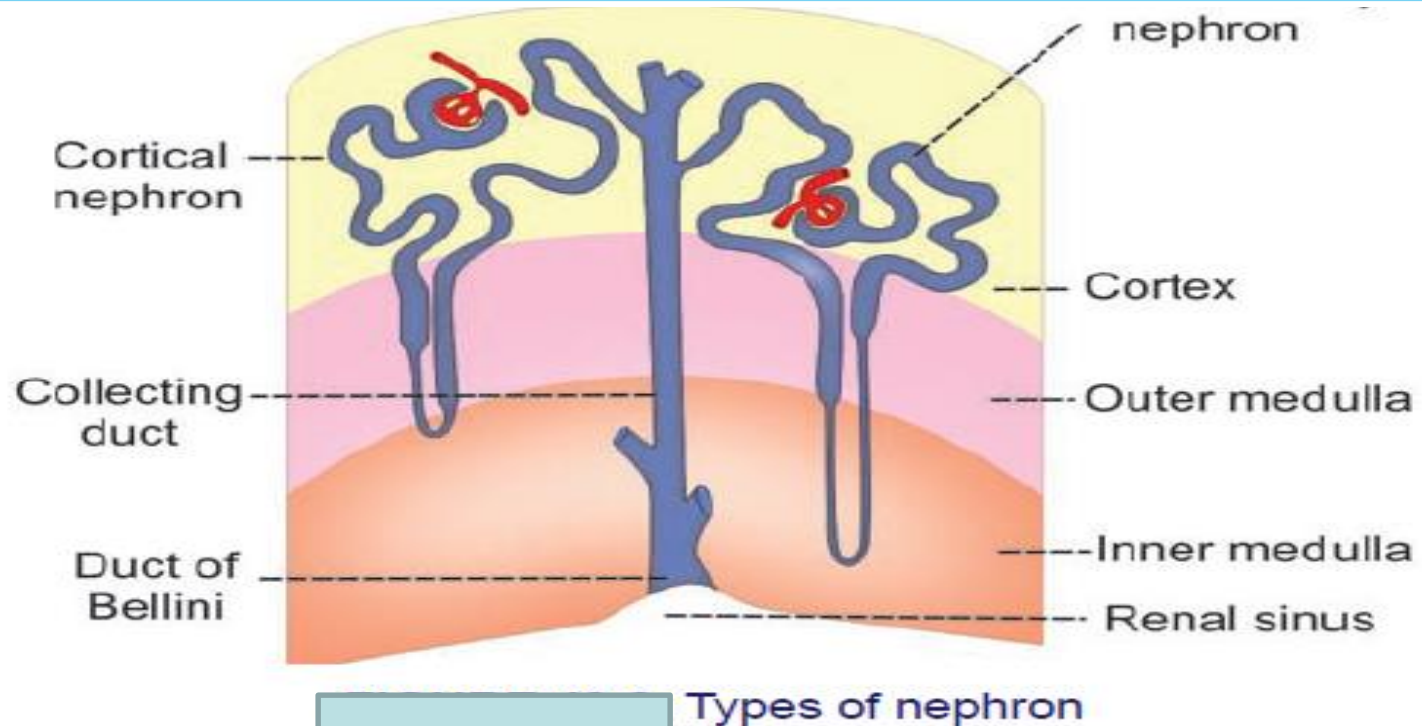
E. Collecting duct

- Is site of ADH activity



2-types of Nephron (Cortical + juxtamedullary)

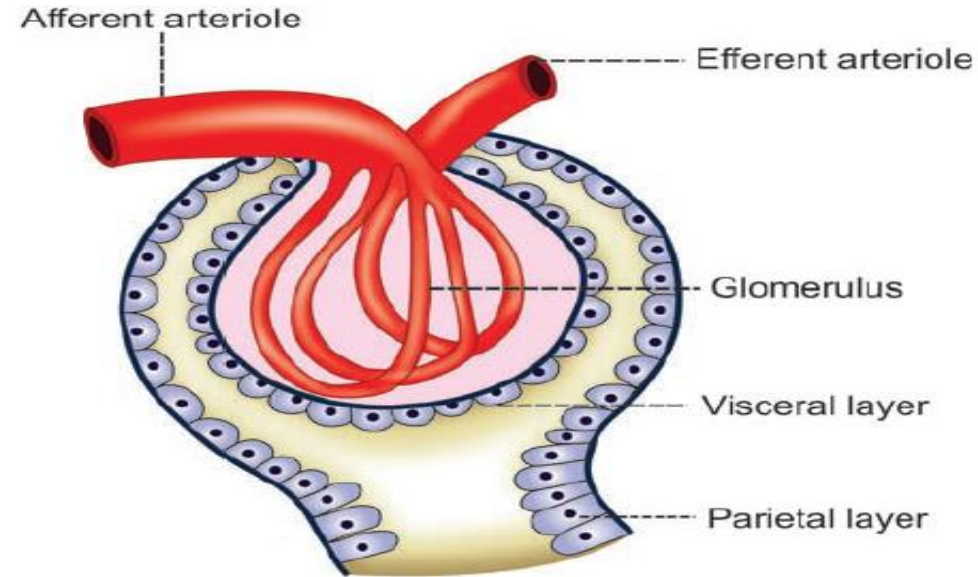
Features	Cortical nephron	Juxtamedullary nephron
Percentage	85%	15%
Situation of renal corpuscle	Outer cortex near the periphery	Inner cortex near medulla
Loop of Henle	Short	Long
	Hairpin bend penetrates only up to outer zone of medulla	Hairpin bend penetrates up to the tip of papilla
Blood supply to tubule	Peritubular capillaries	Vasa recta
Function	Formation of urine	Mainly the concentration of urine and also formation of urine



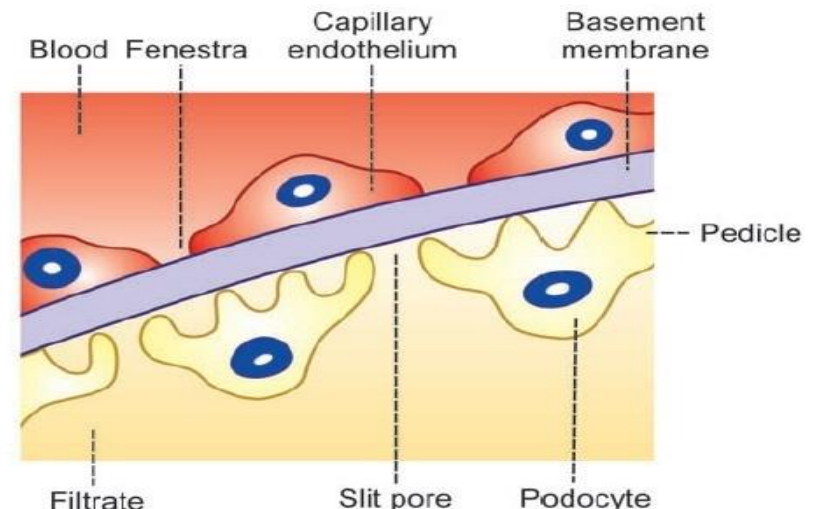
Structure of the glomerulus, site of filtration

3-layers of the filtration membrane

- Glomerular capillary endothelial layer with fenestrae (pores).
 - Basement membrane
 - Visceral portion of the Bowman's epithelial layer with podocytes.
- ✓ **Podocytes are finger-like processes which produce slits for filtration**



Renal corpuscle

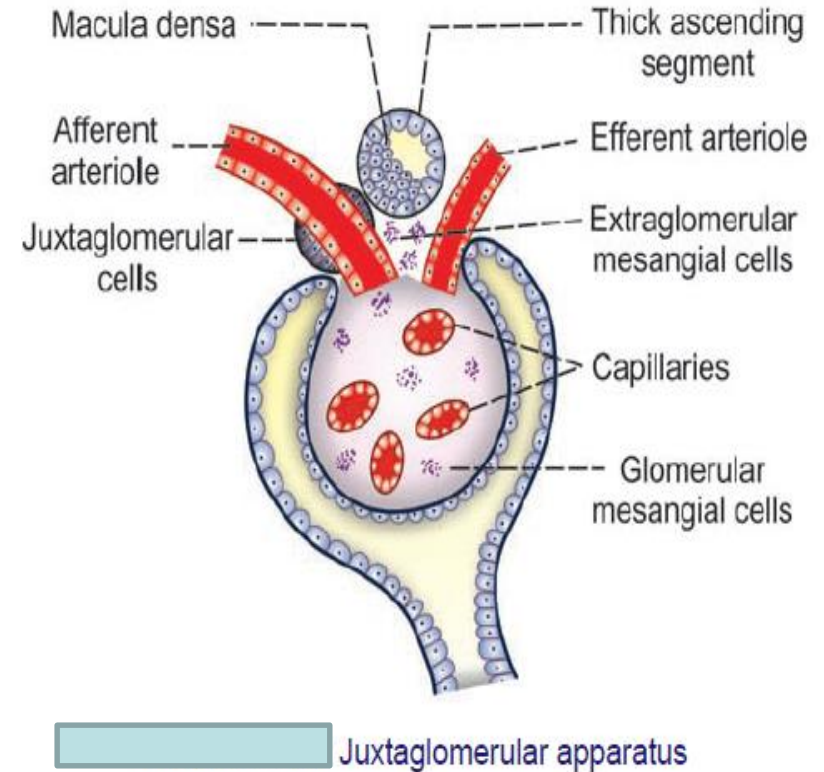


Juxtaglomerular apparatus(JGA) and its function

- ✓ **JGA** is a specialized organ situated near the glomerulus of each nephron
- ✓ It is formed by 3 different structures
 1. Macula densa
 2. Extraglomerular mesangial cells
 3. Juxtaglomerular cells

Function:

- ✓ Secretes hormones like **renin**
- ✓ Regulates **glomerular blood flow** and **GFR**



Renal Autoregulation(AR)

- AR is an **intrinsic (local) mechanism** of the kidney by which GFR and RBF remain constant in conditions of wide fluctuations in ABP (arterial blood pressure)
- Though pressure fluctuates b/n **80 to 170 mm Hg**, GFR and RBF remain more or less constant
- The main function of autoregulation in the kidneys is **to maintain a relatively constant GFR and renal blood flow**

Mechanisms of Auto-regulation

❖ Two mechanisms are involved in renal autoregulation:

1. Myogenic response(**stress relaxation**)
2. Tubuloglomerular feedback

1. Myogenic Response

- Increase in ABP causes the afferent arteriole to stretch greatly at the beginning
- But , after a while, the **smooth muscles of arterioles contract strongly** and **cause decreases in renal blood flow** and finally in **decrease of GFR**

Cont...

- The mechanism: is that **increased stretch by high ABP** causes **increased Ca^{2+} entry into smooth muscles of arterioles** that in turn **induce heavy constriction of the arteriolar walls**
- If arterial blood pressure decreases, the **tonic contraction** of arterioles **disappears** and as the result, the **arterioles are dilated**

2. Tubuloglomerular feedback

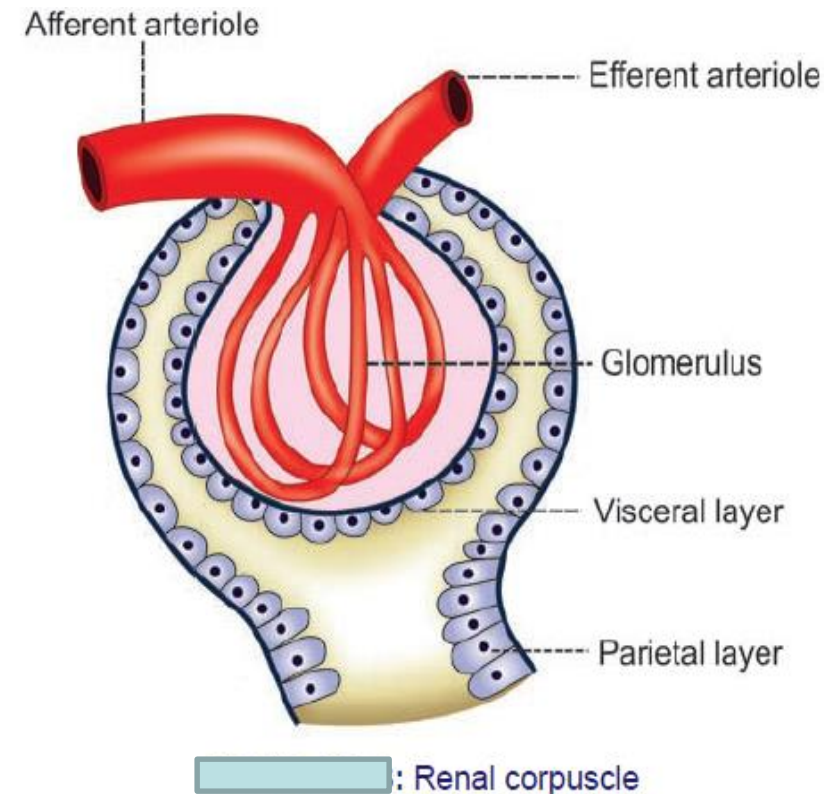
Macula densa - detects level of NaCl in distal tubule and regulates GFR by $[Na^+]$

❖ When there is a **decrease** in **GFR** due to **decrease** in **ABP** then the following effects take place:

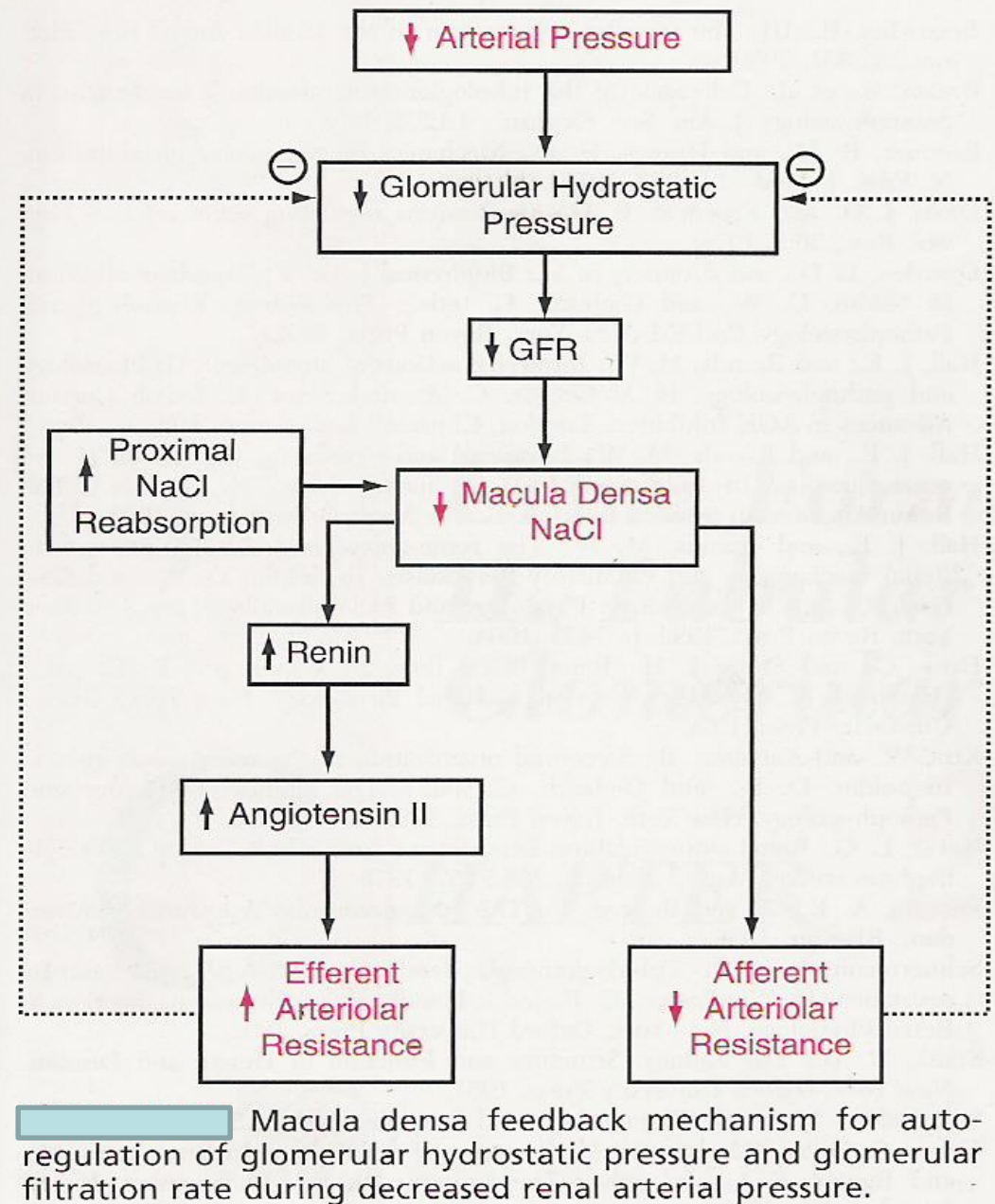
1. Afferent arteriolar feedback

mechanism - causes

vasodilatation of afferent arteriole and thereby **increase GFR**



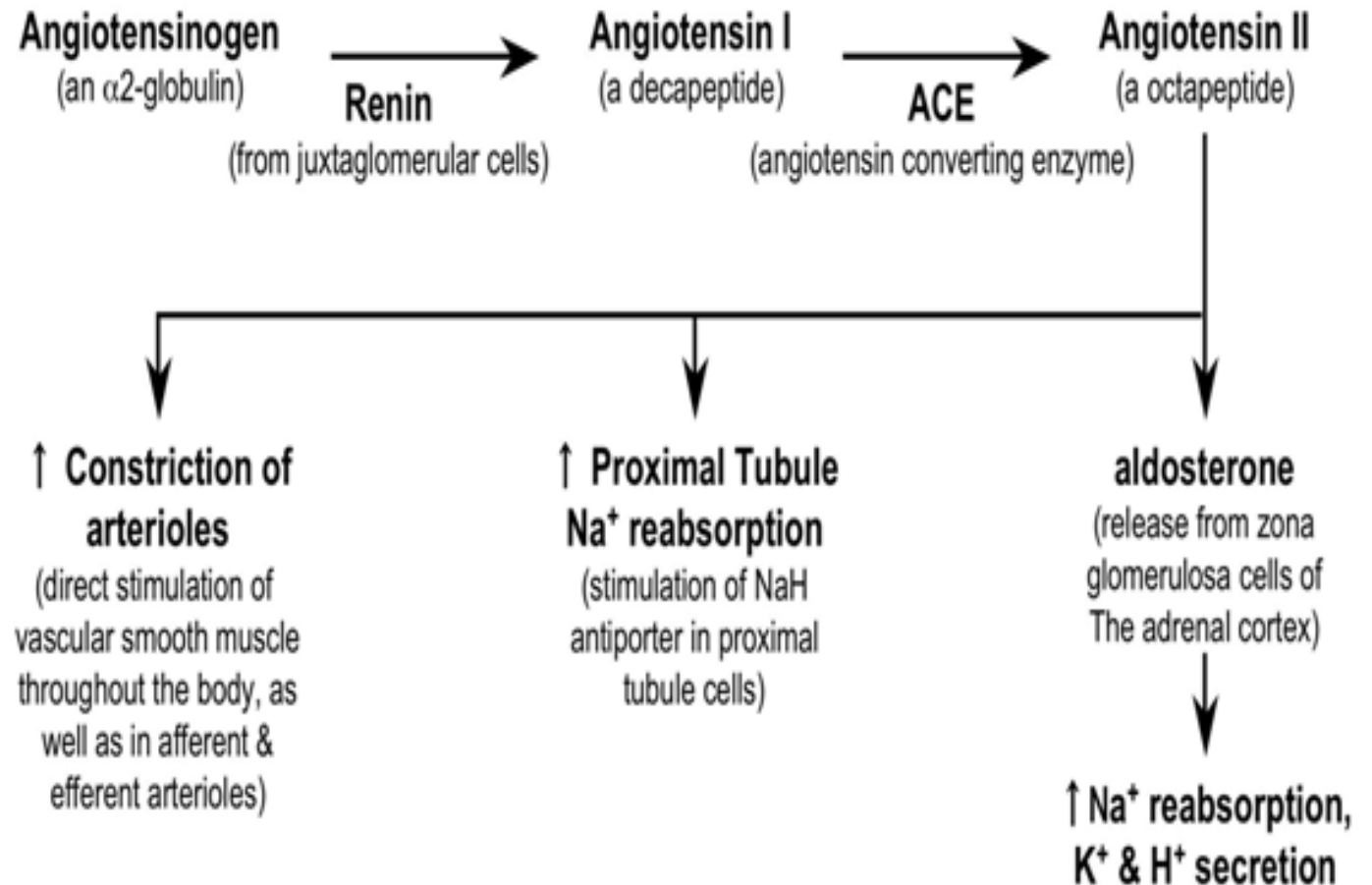
2. Efferent arteriolar feedback mechanism that causes vasoconstriction of efferent arterioles to cause increased GFR
- ✓ This is effected by production of Renin from the kidney that finally forms Angiotensin II



Renin angiotensin aldosterone system (RAAS) in regulation of GFR and ABP

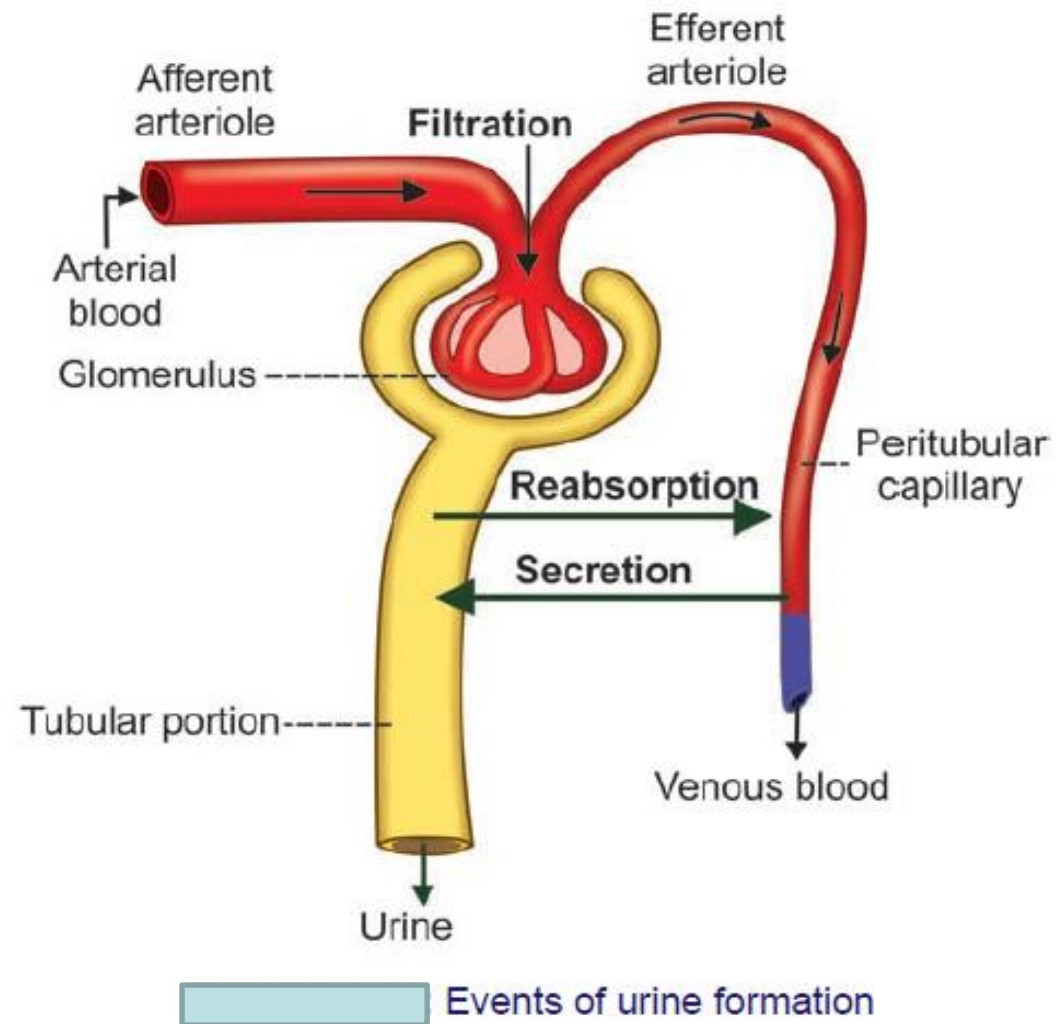
Stimuli for renin secretion

1. Low blood pressure
2. Low ECF volume
3. Sympathetic stimulation
4. Low plasma sodium



Urine formation

- Normally, about 1.3L of blood (26% CO) enters the kidneys.
- Normal urinary output is 1 to 1.5 L/day
- There are 3 processes of urine formation.
 - ✓ Glomerular filtration
 - ✓ Tubular reabsorption
 - ✓ Tubular secretion



Glomerular filtration (GF)

- GF is the process by which the blood is filtered while passing through the glomerular capillaries by filtration membrane.
- It is the first process of urine formation.
- Filtration membrane is formed by three layers:
 - ✓ Glomerular capillary membrane
 - ✓ Basement membrane
 - ✓ Visceral layer of Bowman capsule.
- All the substances of plasma are filtered except the plasma proteins.

Con't

❖ Glomerular filtration rate (GFR)

- is defined as the total quantity of filtrate formed in all the nephrons of both the kidneys in the given unit of time
- Normal GFR is 125 mL/minute or about 180 L/day

Pressures determining GFR

1. Glomerular Capillary Pressure

- Glomerular capillary pressure is the pressure exerted by the blood in glomerular capillaries.
- **It is high pressure (60 mm Hg)** which is important to filter substances to the tubules.

2. Colloidal Osmotic Pressure/ plasma oncotic pressure

- It is the pressure exerted by plasma proteins in the glomeruli.
- It is about **32 mm Hg**.
- It **opposes** glomerular filtration

3. Hydrostatic Pressure in Bowman Capsule

- It is the pressure exerted by the **filtrate** in Bowman capsule.
- It is about **18 mm Hg**. It also **opposes** glomerular filtration

4) Bowman's capsule Oncotic pressure = 0 mmHg under normal conditions

Net Filtration Pressure

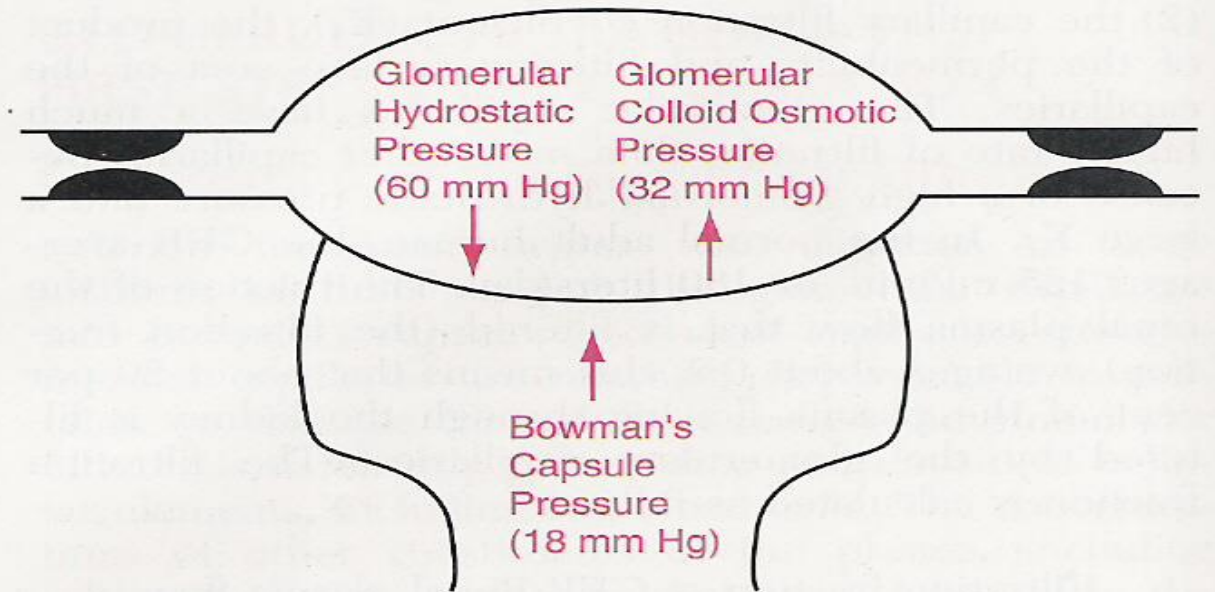
- Net filtration pressure is the **balance** between **pressure favoring filtration** and **pressures opposing filtration**

Cont...

**NFP = pushing
minus Pulling**

$$\text{NFP} = 60 - (32 + 18) \\ = 10 \text{ mm Hg}$$

- **This 10 mm Hg cause filtration to occur.**
- **The filtrate joins bowman's capsule and then to proximal tubule.**
- **Edema can be caused by disturbances of the above variables.**



	Glomerular		Bowman's		Glomerular
	Hydrostatic		Capsule		Oncotic
	Pressure	–	Pressure	–	Pressure
Net Filtration Pressure =	(60 mm Hg)		(18 mm Hg)		(32 mm Hg)

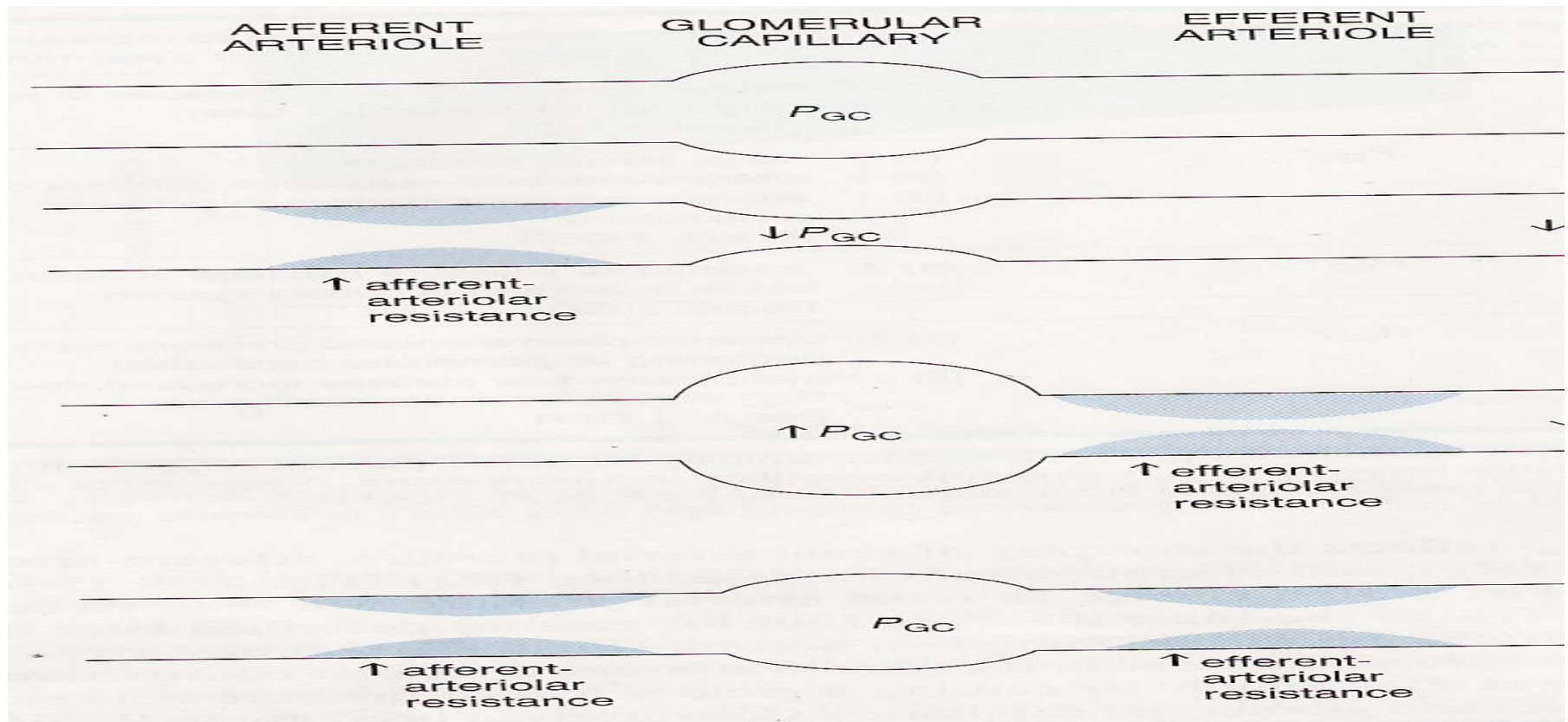
Summary Summary of forces causing filtration by the glomerular capillaries. The values shown are estimates for normal humans.

Factors regulating (affecting) GFR

- *Renal Blood Flow*
 - *Glomerular Capillary Pressure*
 - *Surface Area of Capillary Membrane*
 - *Permeability of Capillary Membrane*
- Directly proportional to GFR
- *Hydrostatic Pressure in Bowman's Capsule & Colloidal Osmotic Pressure are **inversely** related to **GFR***
 - Increased **Sympathetic** stimulation = Decreases GFR
 - Renal blood flow is decreased during **exercise** and **heavy stress**

Con't

- **Afferent arteriolar constriction** : Decreases GFR
- **Efferent arteriolar constriction**: increased GFR. Constriction of the efferent arterioles increase resistance to blood flow and this effect results in increased GFR.



Tubular reabsorption

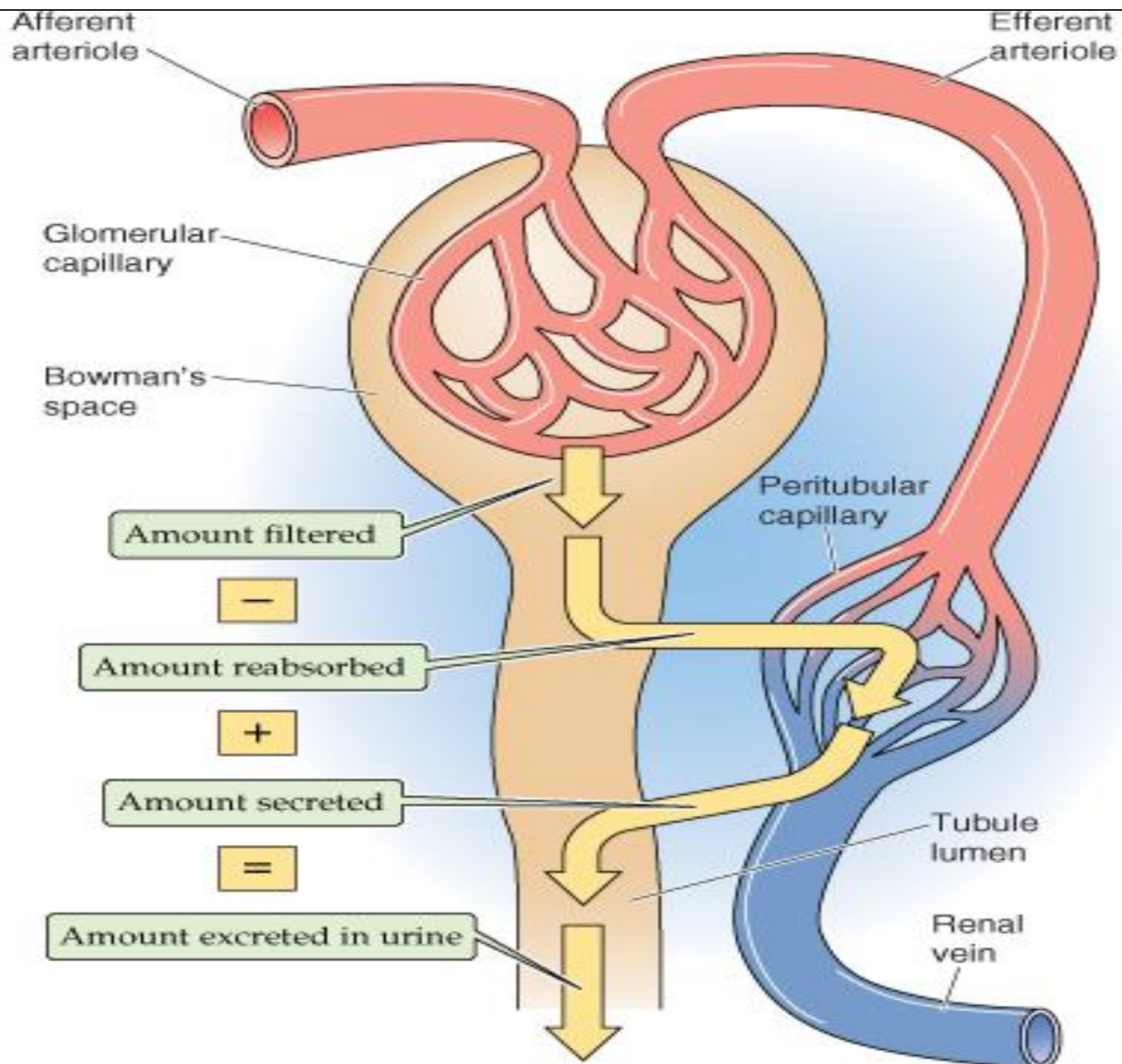
- ✓ Process of transporting water and other substances from renal tubules back to the blood

- ✓ The tubular cells reabsorb only the substances necessary for the body.

178.5 L/day (99%) of filtrate reabsorbed

1.5 L/day (<1%) lost as urine

- ✓ The reabsorbed substances move into the interstitial fluid of renal medulla. And, from here, the substances move into the blood in peritubular capillaries



Cont...

- ✓ Oncotic pressure is greater than hydrostatic pressure in peritubular capillaries and this makes Reabsorption to be easier
 - Reabsorb 99% H₂O, 100% glucose, 99.5% Na⁺ and 50% urea.
 - Most of this occurs at proximal convoluted tubule

Mechanism of reabsorption

- Basic transport mechanisms involved in tubular reabsorption are of two types:
 1. Active reabsorption
 2. Passive reabsorption
- Reabsorption of substances from **tubular lumen** into the **peritubular capillary** occurs by **two routes**:
 1. Transcellular route
 2. Paracellular route

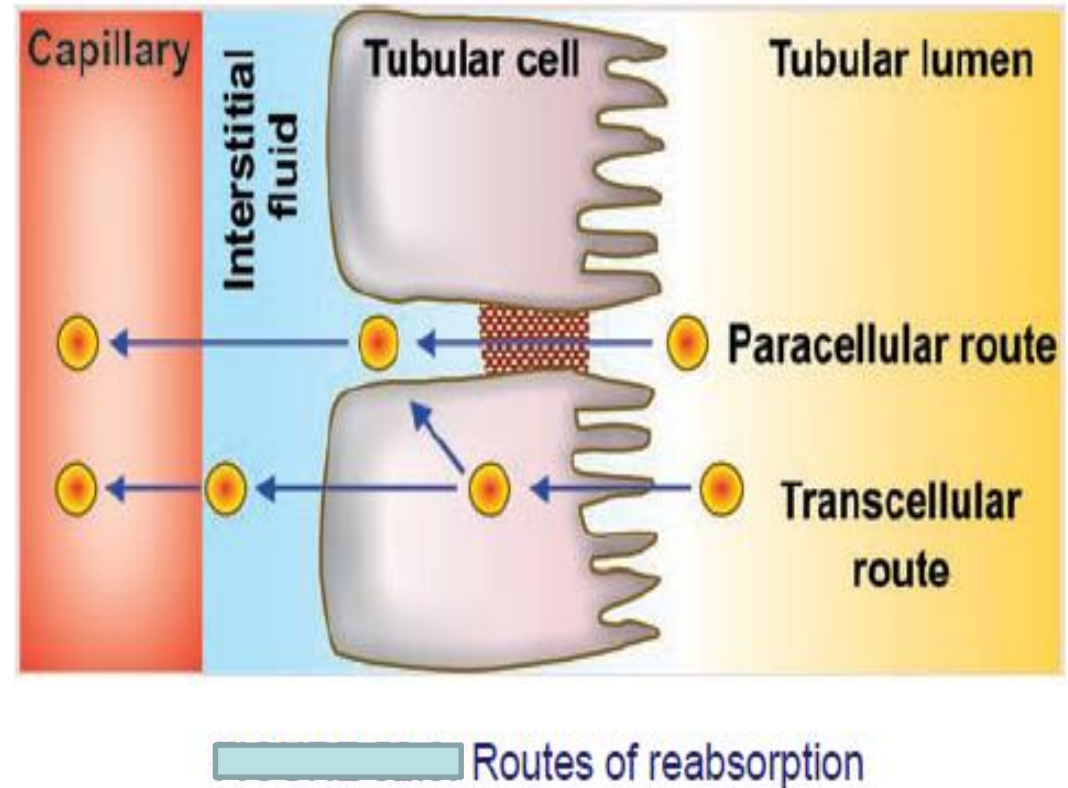
Con't

1. Transcellular route:

passing *through* luminal epithelial cells or basolateral membrane (**mostly active transport**)

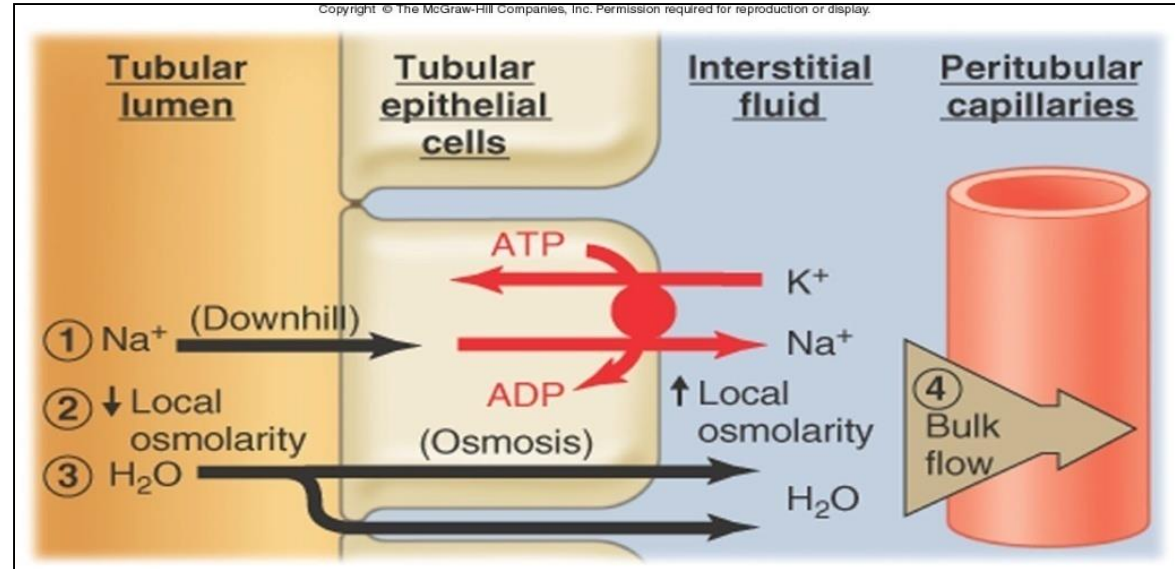
2. Para-cellular route:

- Passing *between* tight junctions or lateral spaces (**mostly diffusion**)



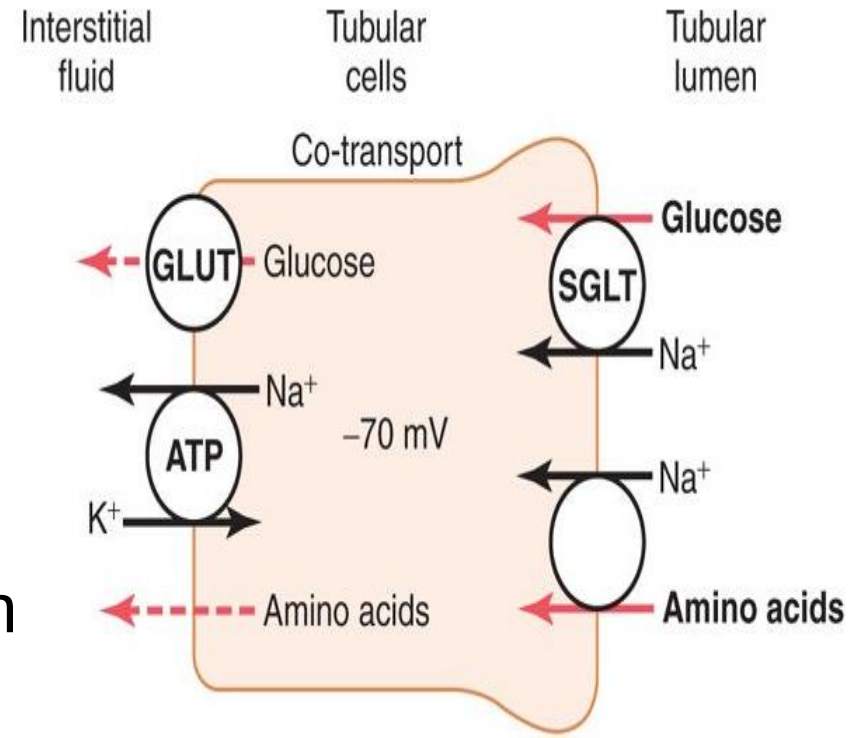
1) Na⁺ transport

- Na⁺ absorbed by active transport mechanisms.
- Basolateral ATPase's establish a gradient across the tubule wall.
- 2/3 of Na⁺ reabsorbed in PCT
- remaining 1/3 reabsorbed in other segments.
- H₂O follows Na⁺ due to osmotic force



2) Glucose transport

- Completely reabsorbed in the PCT.
- absorption relies upon Na^+ .
- At **apical membrane** - needs Na^+ /glucose co-transporter (**SGLT**)
- At **basolateral membrane** - via glucose transporter (**GLUT's**), which does not rely upon Na^+ .



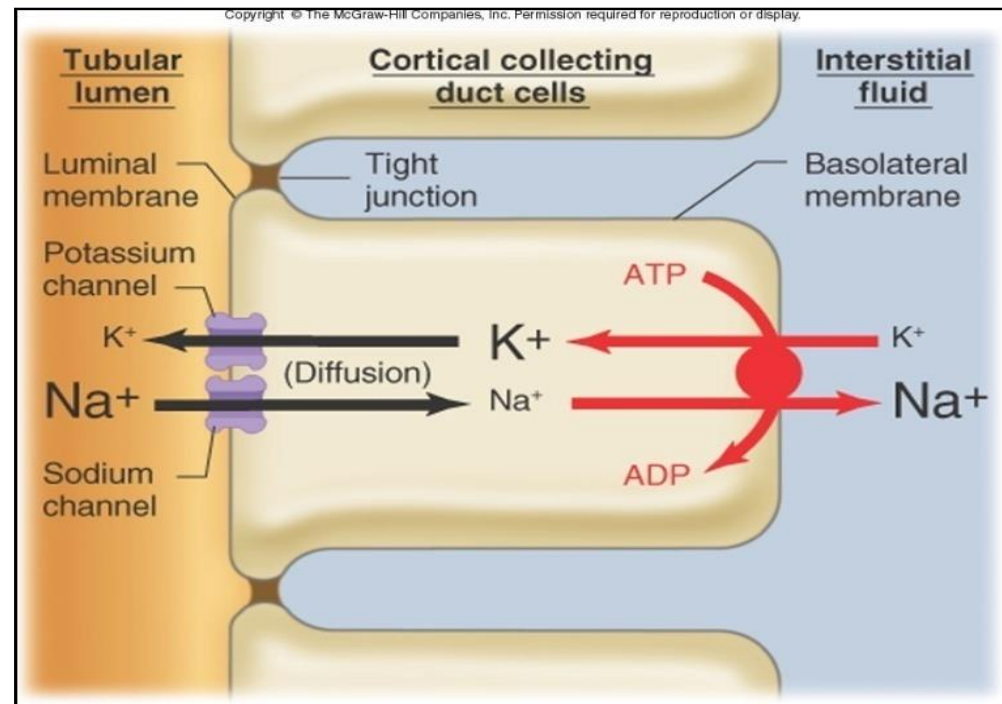
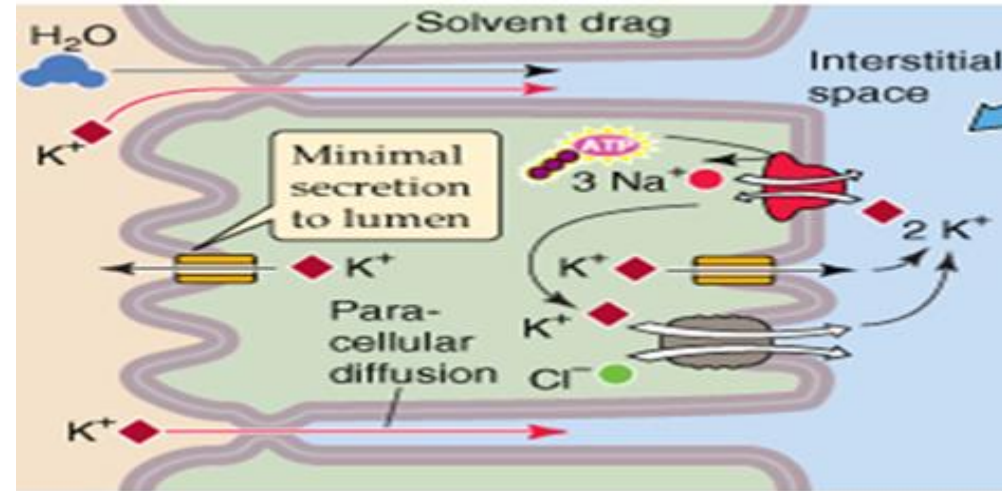
3) Amino acid transport

- ✓ **Rely upon Na^+ gradient at apical membrane**

4) K^+ transport

- K^+ Reabsorption along the **PT** is largely passive and follows the movement of Na^+ and fluid.
- K^+ secretion occurs in **CT** and relies upon active transport of K^+ across basolateral membrane and passive exit across apical membrane into tubular fluid.

At Proximal tubule(**PT**)



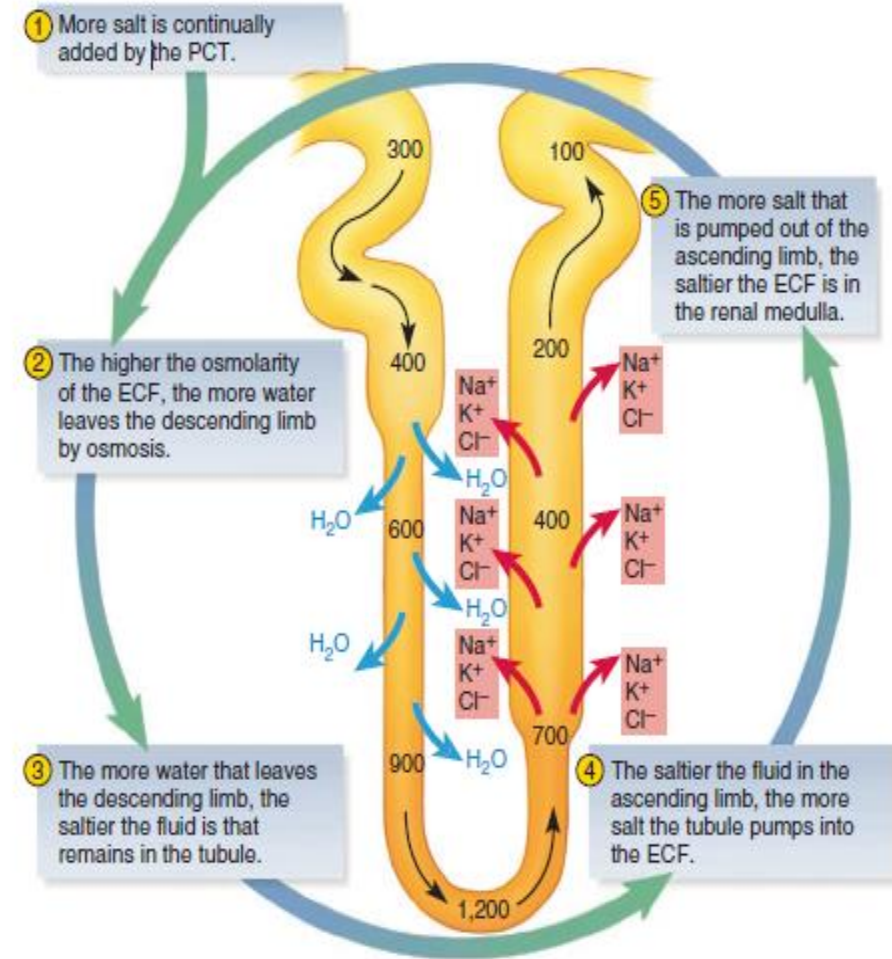
5) H₂O absorption in Renal tubules

In the proximal part:

- H₂O reabsorption is mainly coupled to Na⁺ reabsorption in proximal tubule.
- H₂O also drags some solutes with it by the so called **solvent drag**

In the loop of Henley:

- **Descending portion - is H₂O permeable.**
- **Ascending loop - is not H₂O permeable**, but **only permeable for solutes like Na⁺, Cl⁻, K⁺, etc**



In the distal and collecting ducts:

- H_2O reabsorption fully depends on the presence or absence of ADH.
- Principal cells: responsible for Na^+ reabsorption and K^+ secretion
- Intercalated cells: responsible for K^+ and HCO_3^- reabsorption and H^+ secretion.

Reabsorption processes in the tubules

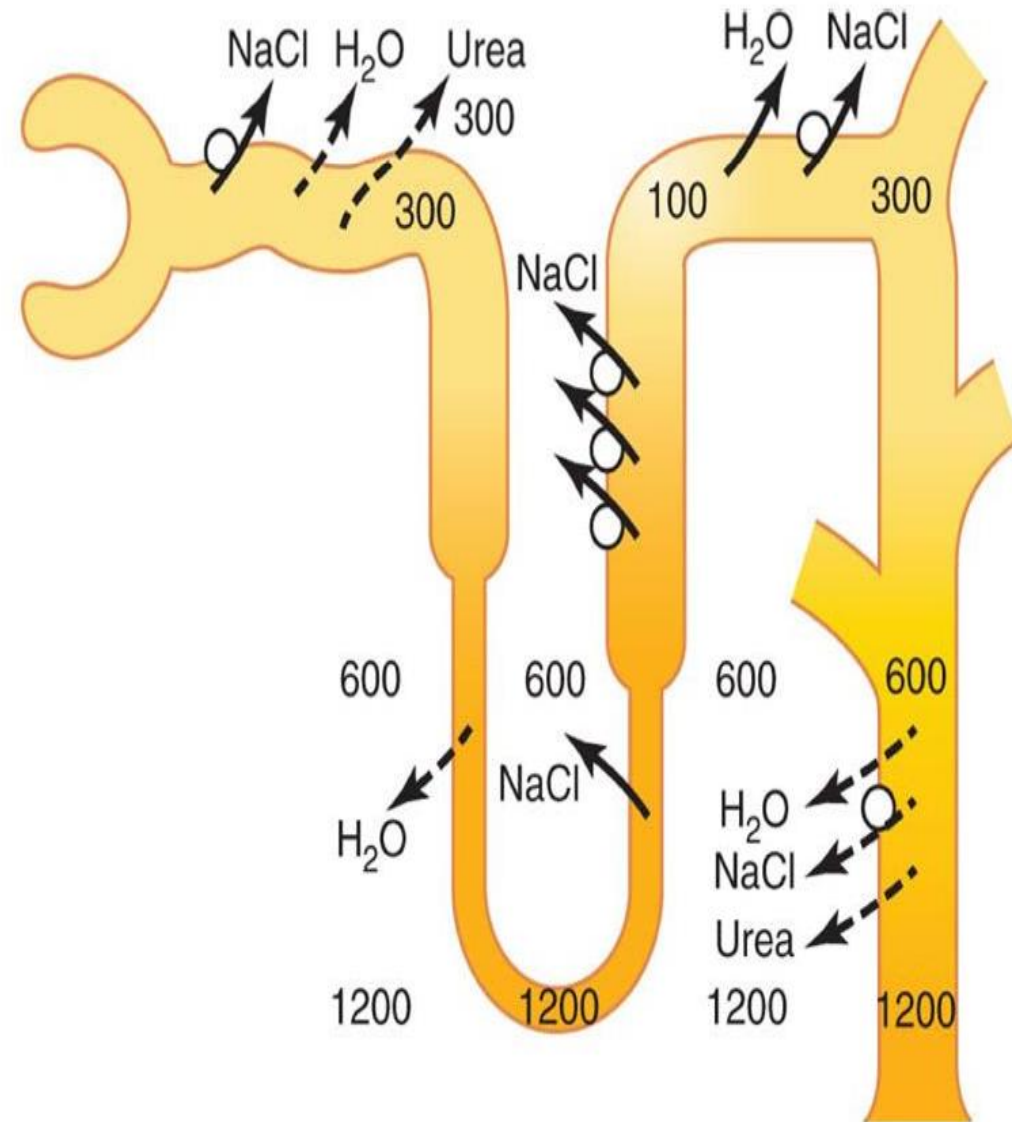
Generally the reabsorption process in different tubular segments are the following.

1) In the proximal tubule

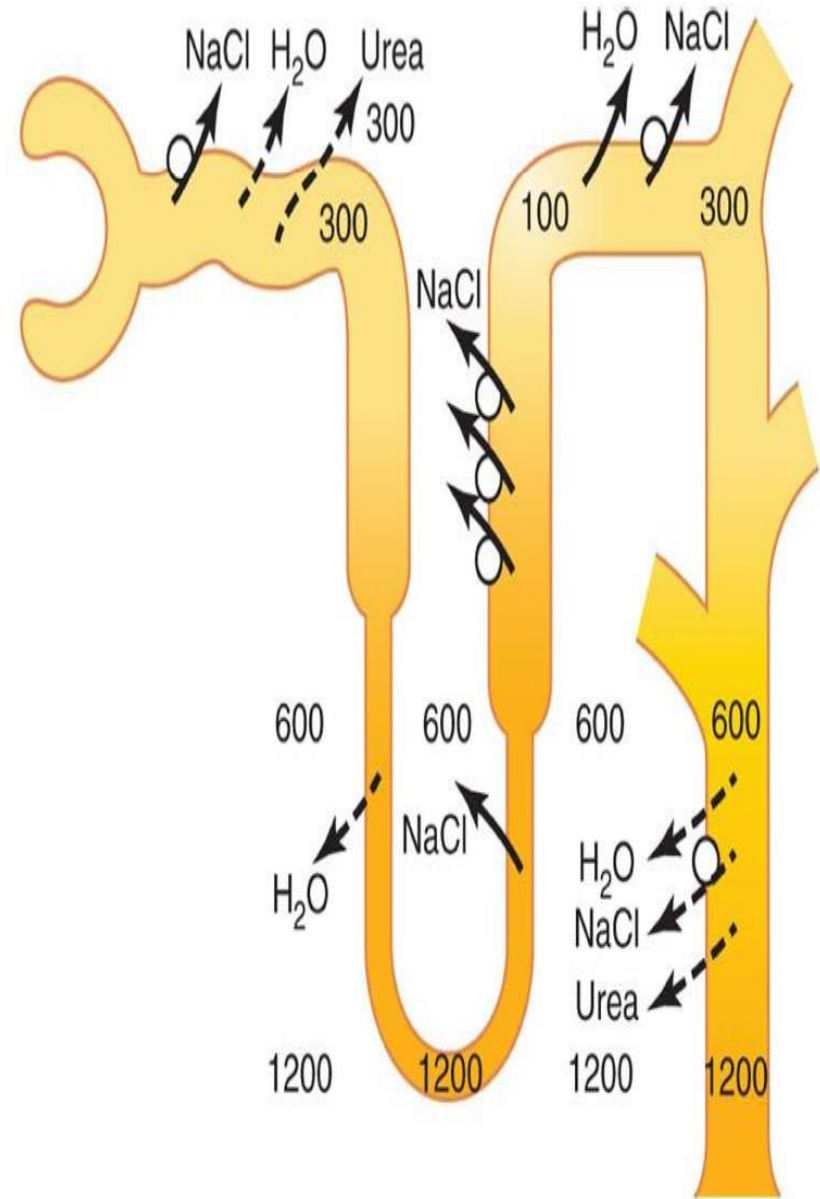
- **About 65% of the filtrate is reabsorbed.**
 - **These includes: electrolytes like Na^+ , K^+ , Cl^- , etc;**
 - **Moreover, organic substances like, glucose & amino acid are completely reabsorbed.**

2) Reabsorption of the filtrate in the Loop of Henley

- Descending limb
 - At the descending limb H_2O escapes to the **interstitium** from the **lumen** and creates hyperosmolar environment inside the lumen to a level of **1200 mOsm/l**



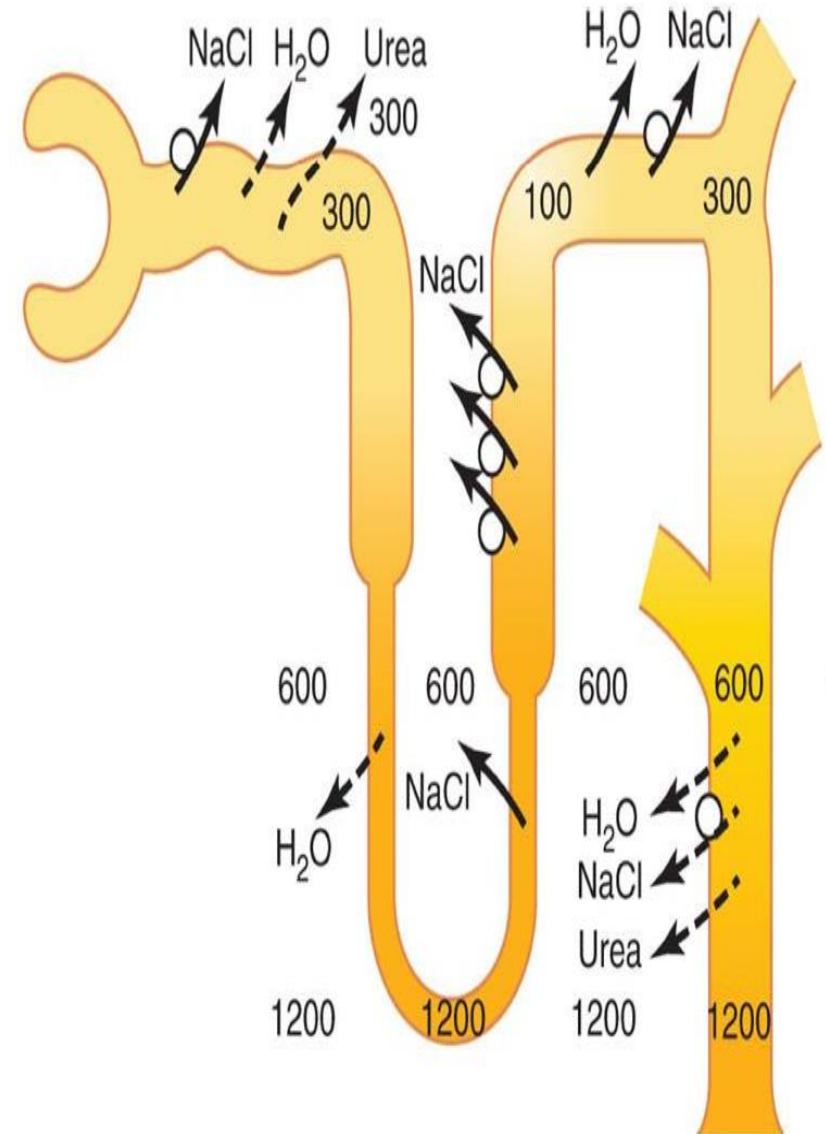
- **Ascending limb:**
 - Is **impermeable to water** but **permeable to solutes**.
 - Transport of NaCl to the interstitium by active transport causes a **hypoosmolar** environment, which is about = **100 mOsm/l**
 - finally the dilute fluid enters the distal tubule from the thick ascending limb



3) Reabsorption of the filtrate in DT and CD

Some 18 to 19% of the filtrate is reabsorbed in the distal and collecting duct.

- Only 1 to 2 liters of the filtrate (1%) is excreted as urine, where **ADH** plays a big role in concentration and dilution of urine



Regulation of tubular reabsorption

Tubular reabsorption is regulated by 2 factors

A) Hormonal

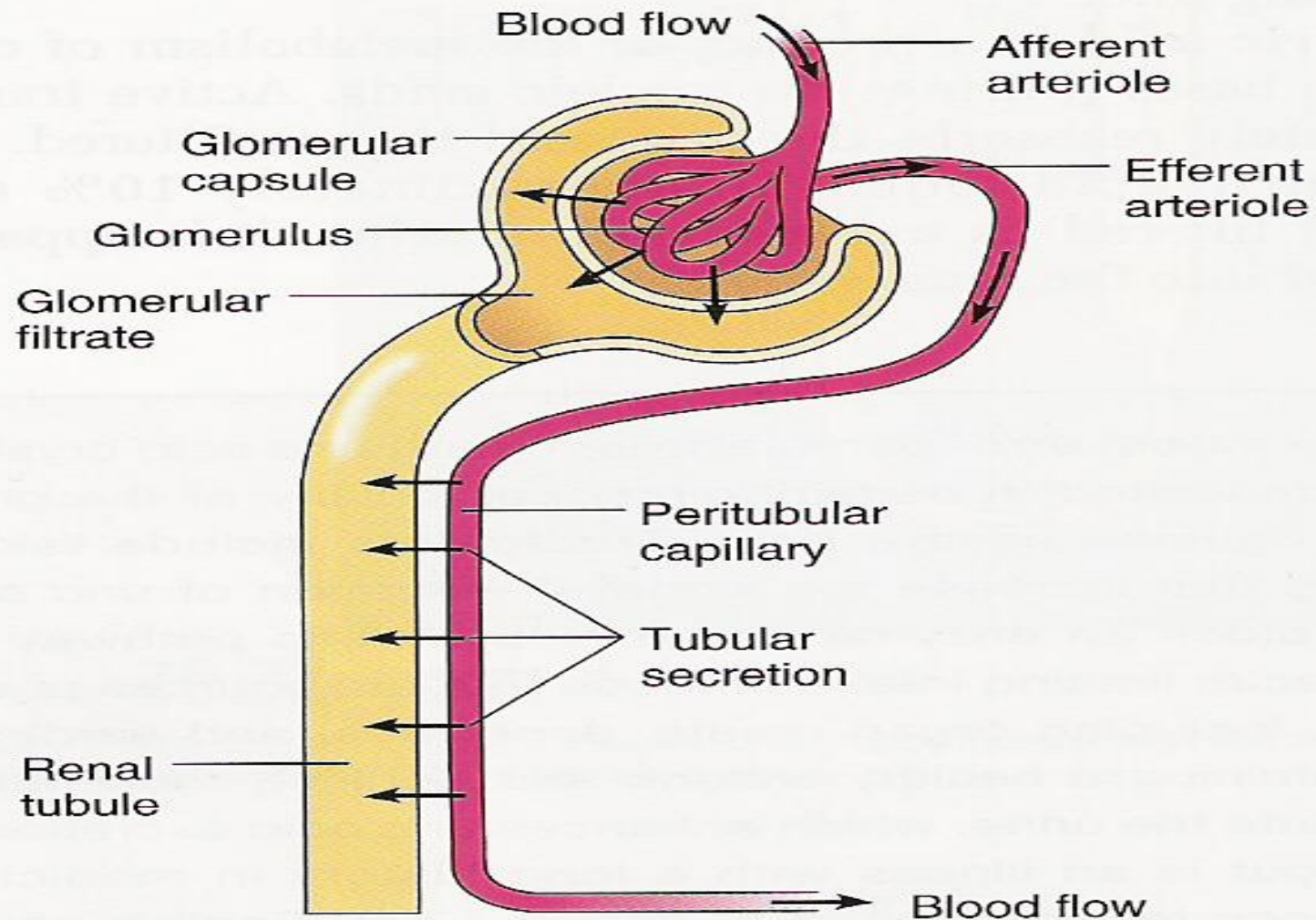
1. **Aldosterone**: **increases Na + reabsorption** and **increases K⁺ secretion**.
2. **Angiotensin II**. **Increases Na + and H₂O absorption (PCT)**
3. **ADH**: **Increases H₂O absorption in distal collecting duct**

B) Nervous

- SNS stimulation increases **Renin** release that in turn enhances **Angiotensin II** formation.

Tubular secretion

- ***Tubular secretion*** is the process by which the substances are transported from the **Peritubular capillaries** and **interstitial fluid** into the tubular lumen
- Toxic or unwanted metabolic by-products like Creatinine, NH_3 , urea, uric acid & some drugs are secreted and excreted
- K^+ , H^+ are also secreted actively mainly from the distal tubule



Secretory mechanisms move substances from the plasma of the peritubular capillary into the fluid of the renal tubule.

Concentration and dilution of urine

- A. Why necessary to excrete a concentrated urine?
- *To save water and thus control ECF- volume during dehydration.*
- B. Why necessary to excrete a dilute urine?
- *To remove excess H_2O (fluid overload) from the body during hyperhydration*
- Generally, concentration and dilution of urine is important to regulate the water and solute (osmolality) level of the ECF-volume.

Mechanisms of concentration & dilution of Urine

❖ *3-important mechanisms are involved in concentration and dilution of urine:*

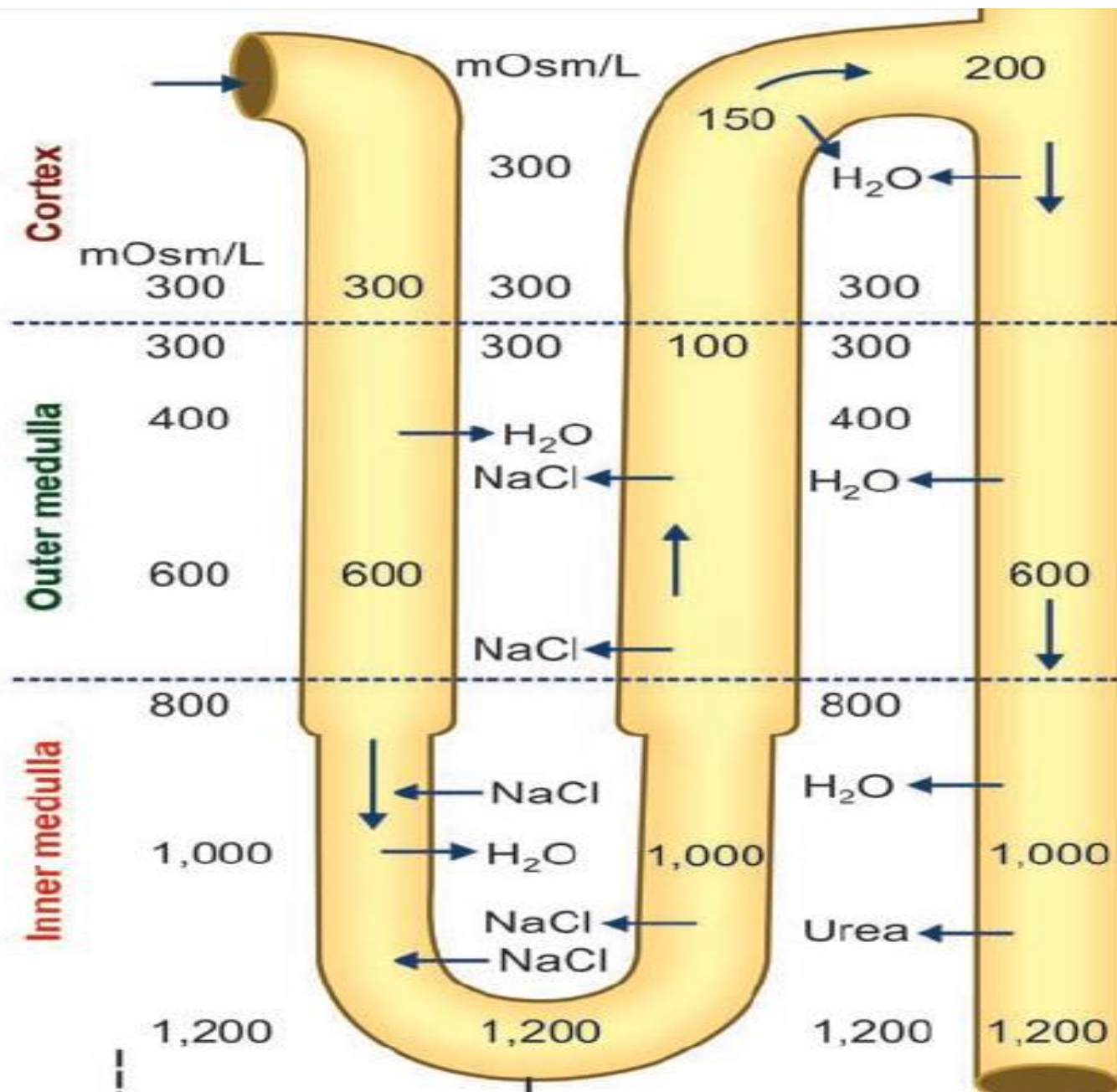
A. Counter current multiplier and counter current exchanger that produce Hyperosmolar medullary interstitium

B. The presence and absence of ADH

C. The importance of urea in increasing the osmolality of the medullary interstitium.

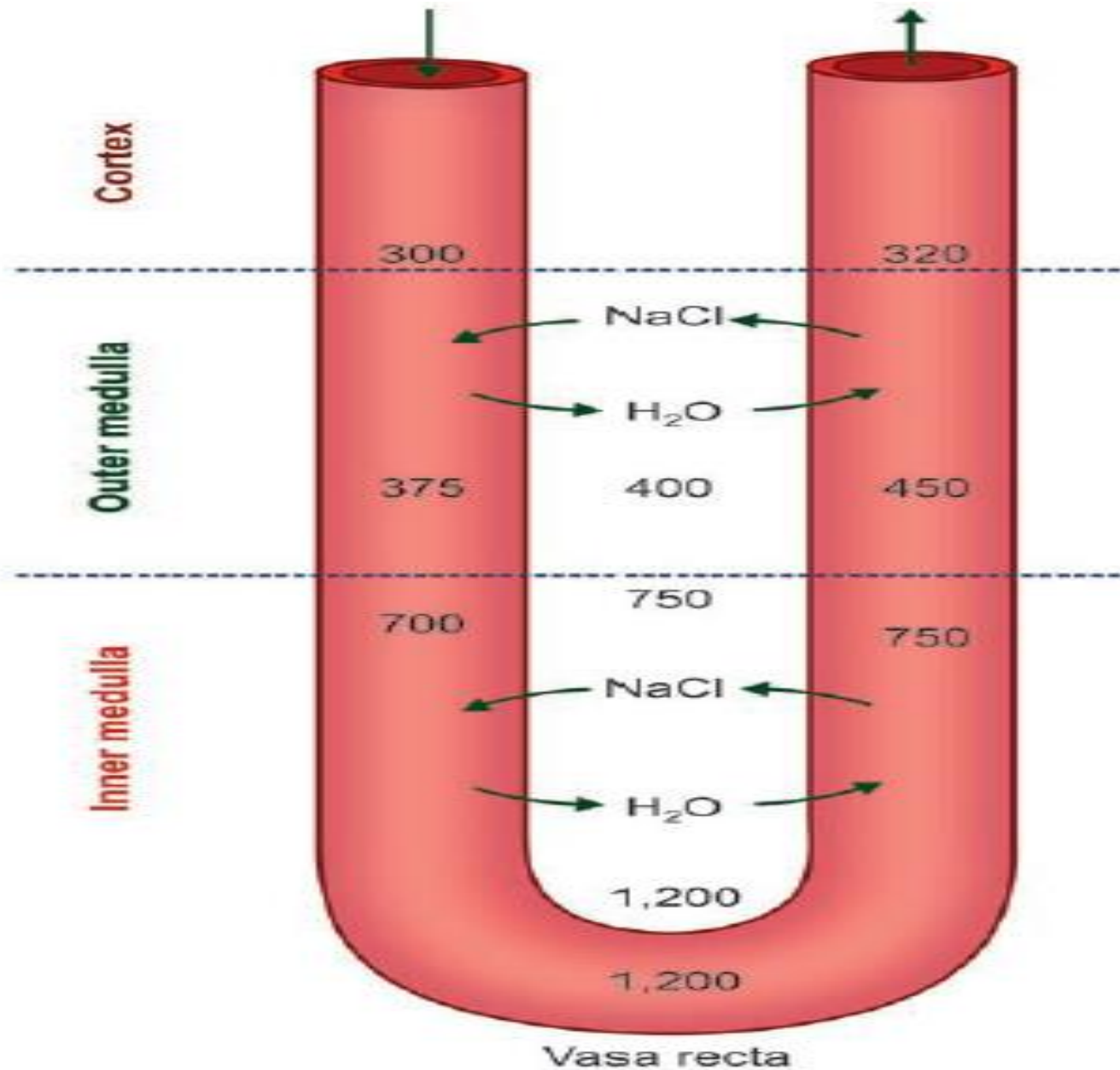
A) Counter current multiplier

- occurs at **Loop of Henley**
 - Solute concentration keeps on increasing in the tubules and forms **hyperosmolar concentration in the Loop of Henley (1200 mOsm/l)**
 - The descending tubule is permeable **only to water**
 - The ascending (thick tubule) is **only permeable** to solutes (Na^+ , Cl^- etc.)
- Result:** The net result of the counter current multiplier system is to produce **hyper-osmotic interstitial fluid in the medulla.**



Counter current exchanger

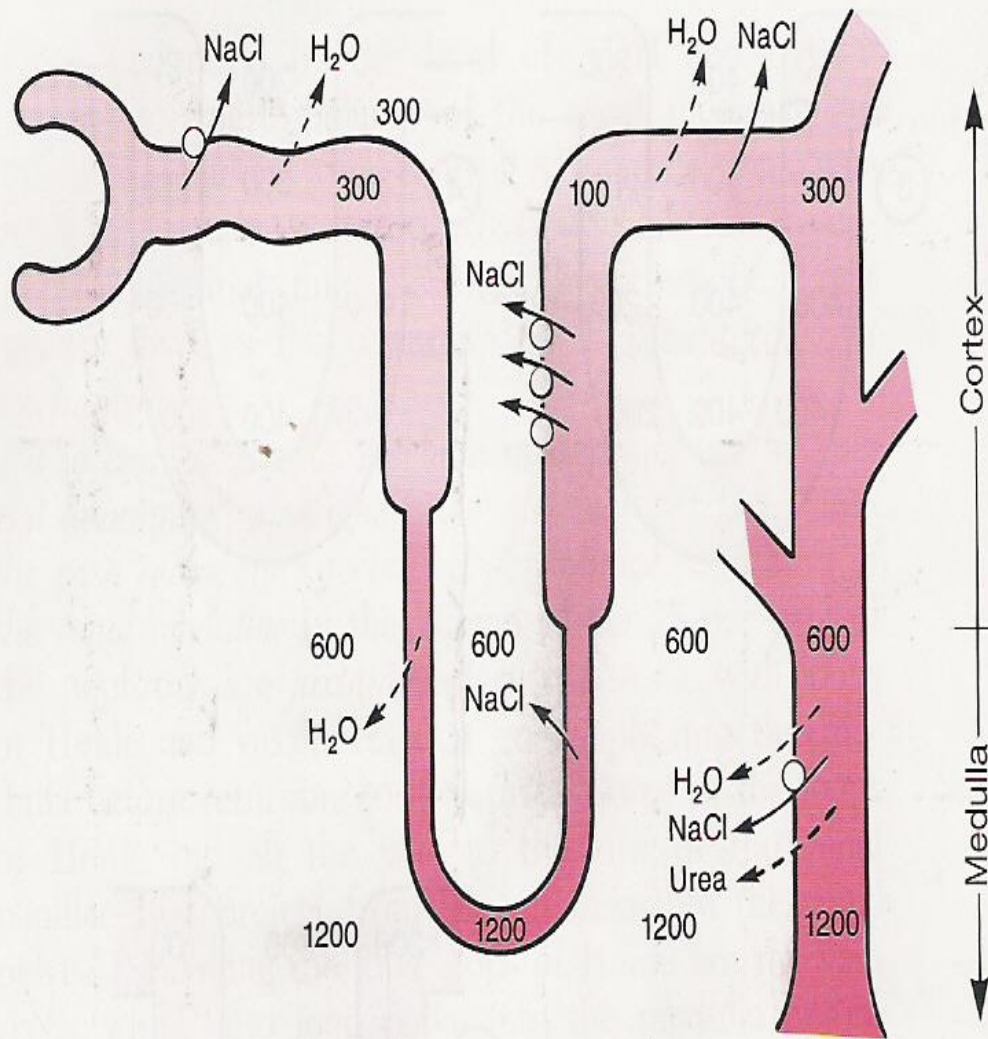
- Occurs by **vasa recta**
- It prevents washout or removal of solutes (salts) from medullary interstitium
- It is responsible for the **maintenance of medullary gradient**, which is developed by **countercurrent multiplier**
- **NaCl and urea** diffuse out of the ascending limb and **enter into the descending limb.**
- **H₂O** diffuses out of the descending limb and enter into the ascending limb



Counter current exchanger.
Numerical indicate osmolarity (mOsm/L)

B) Role of ADH

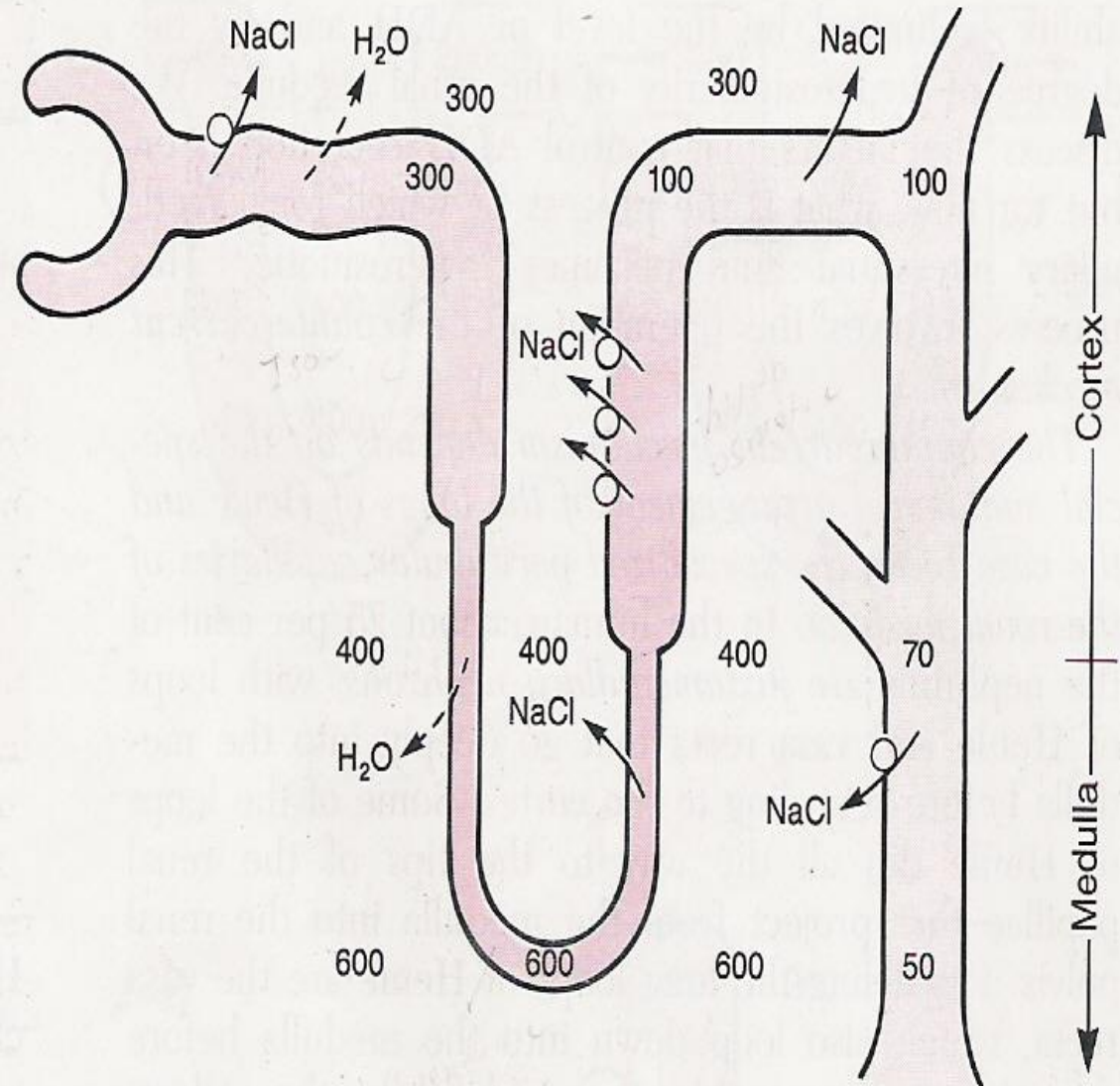
Formation of a concentrated urine when ADH is high



Formation of a concentrated urine when ADH levels are high. Note that the fluid leaving the loop of Henle is dilute but becomes concentrated as water is absorbed from the distal tubules and collecting tubules. With high ADH levels, the osmolarity of the urine is about the same as the osmolarity of the renal medullary interstitial fluid in the papilla, which is about 1200 mOsm/liter. (Numerical values are in milliosmoles per liter.)

Formation of dilute urine when ADH is low

-2. Formation of a dilute urine when ADH is very low. Note that in the ascending loop of the tubular fluid becomes very dilute. In the distal and collecting tubules, the tubular fluid is diluted by the reabsorption of sodium chloride. The failure to reabsorb water when ADH levels are low. The failure to reabsorb water and continued reabsorption of solutes lead to a large volume of dilute urine. (Numerical values are in milliosmoles per liter.)



C) Urea increases the Osmolarity of the medullary interstitium

- The hyperosmolarity of the medullary Interstitium is **not** only due to **NaCl and K⁺**, but also **due to urea**.
- **Urea** is transported (**facilitated diffusion using Urea transporters**) from **collecting duct** and accumulates in the medulla to cause the hyperosmolar environment.
- **A high-protein diet** increases the ability of the kidneys to **concentrate urine**.

H₂O reabsorption

- Facultative:
 - H₂O is reabsorbed with the presence of ADH (in collecting duct).
- Obligatory:
 - H₂O reabsorption takes place without the presence of ADH (proximal tubule)

Stimulation of ADH on collecting duct cells

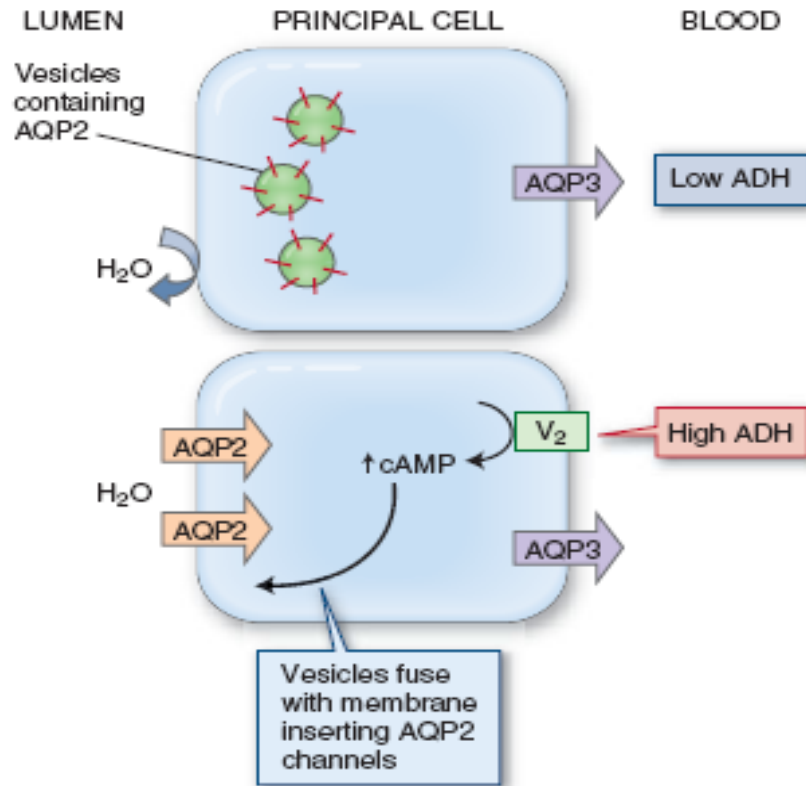
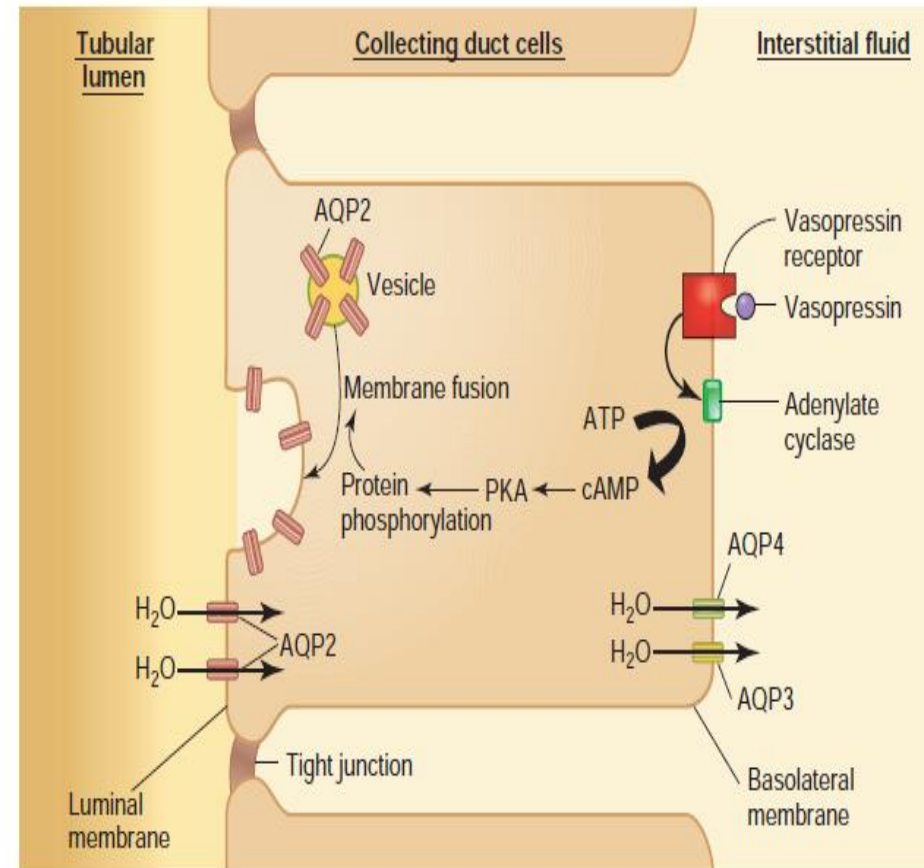


Figure 6-27: Aquaporin 2 (AQP 2) water channel expression in the collecting duct. In the absence of antidiuretic hormone (ADH), the water permeability in the collecting duct is low because water channels are not present in the apical membranes. AQP3 channels are expressed continually in the basolateral membrane. Binding of ADH to the V_2 receptors on the basolateral membrane of the principal cells causes insertion of vesicles containing AQP2 water channels into the apical cell membrane, completing the pathway for water flux.

- Because ADH is a hormone that moves with the blood, it's receptors (V_2 -receptors) are found on the **basolateral side** of the epithelial cells of the collecting duct.
- Stimulation with increased **ADH** causes aquaporin (water channels) to move to luminal side to allow water osmosis

- Vasopressin binding to its receptor increases intracellular **cAMP**
- cAMP increases the activity of the enzyme **PKA**
- PKA increases the phosphorylation of specific proteins that increase the rate of the fusion of vesicles (containing AQP2) with the luminal membrane.
- This leads to an increase in the number of AQP2 channels in the luminal membrane.
- This allows increased passive diffusion of water into the cell.
- Water exits the cell through AQP3 and AQP4 (to interstitial fluid) which are not vasopressin sensitive



Regulation of ADH

Increase ADH

- ↑ plasma osmolality
- ↓ blood volume
- ↓ blood pressure
- Drugs like Morphine, Nicotine, Cyclophosphamide

Decrease ADH

- ↓ Plasma osmolality
- ↑ Blood volume
- ↑ Blood pressure
- Drugs like alcohol, clonidine(antihypertensive drug) , haloperidol(dopamine blocker)

Disorders of urinary concentrating ability: 2-types of diseases

a. Central (brain) D. insipidus

- low ADH secretion from the hypothalamic brain due to:-
 - i. Lesion (injury) of **SO** and **PV** nuclei of hypothalamus
 - ii. Lesion in hypothalamo-hypophyseal tract
 - iii. Atrophy of posterior pituitary

b. Nephrogenic (kidney) D. insipidus

- when the distal and collecting tubule duct **receptors do not respond** to **ADH**

Ureters

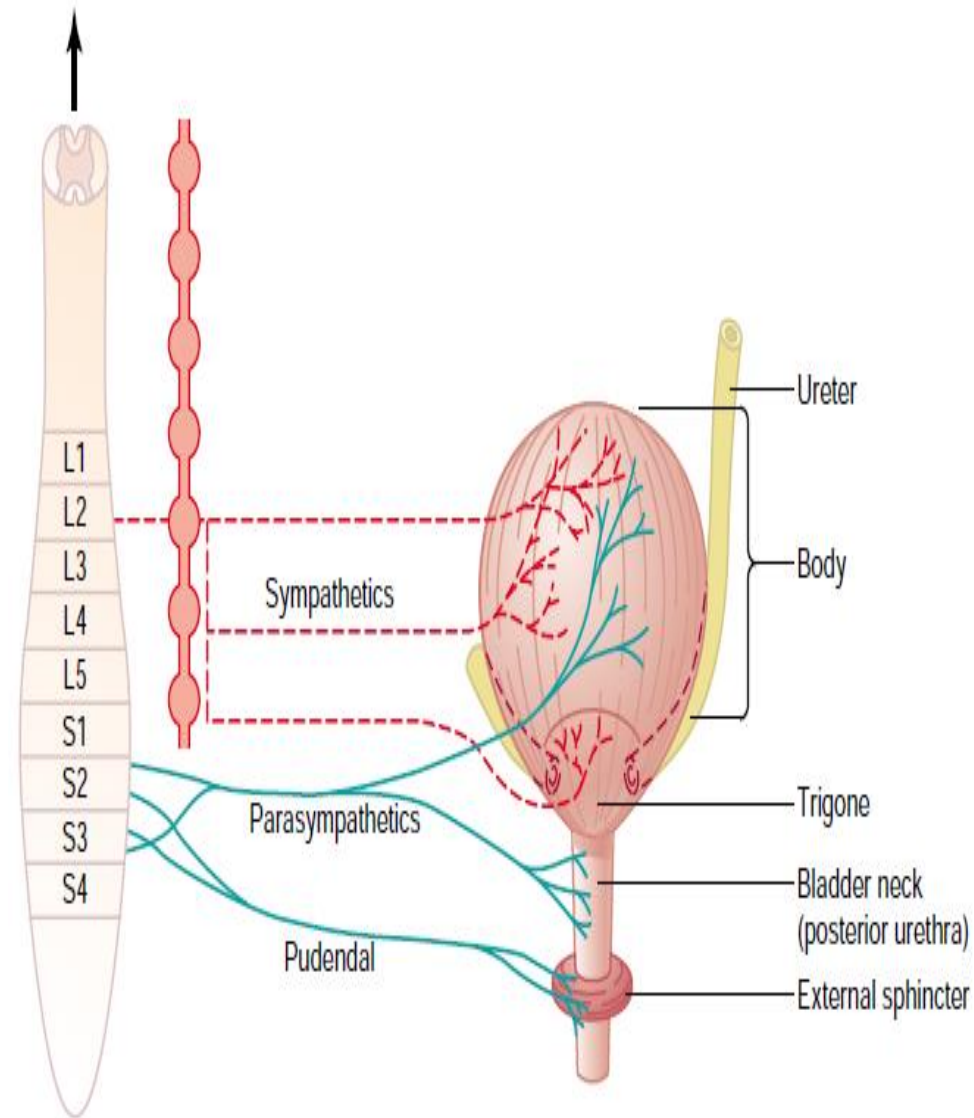
- Slender tubes that convey urine from the kidneys to the bladder
- Ureters enter the base of the bladder through the posterior wall
- Ureters actively propel urine to the bladder via response to smooth muscle stretch

Urinary bladder and its innervation

- Bladder wall is made up of smooth muscle called **detrusor muscle**
- Is distensible and collapsible that temporarily stores urine
- Accommodates as high as 1.5L of urine

Innervated by:

- **SNS**
- **PNS**
- **Somatic nerve fiber**

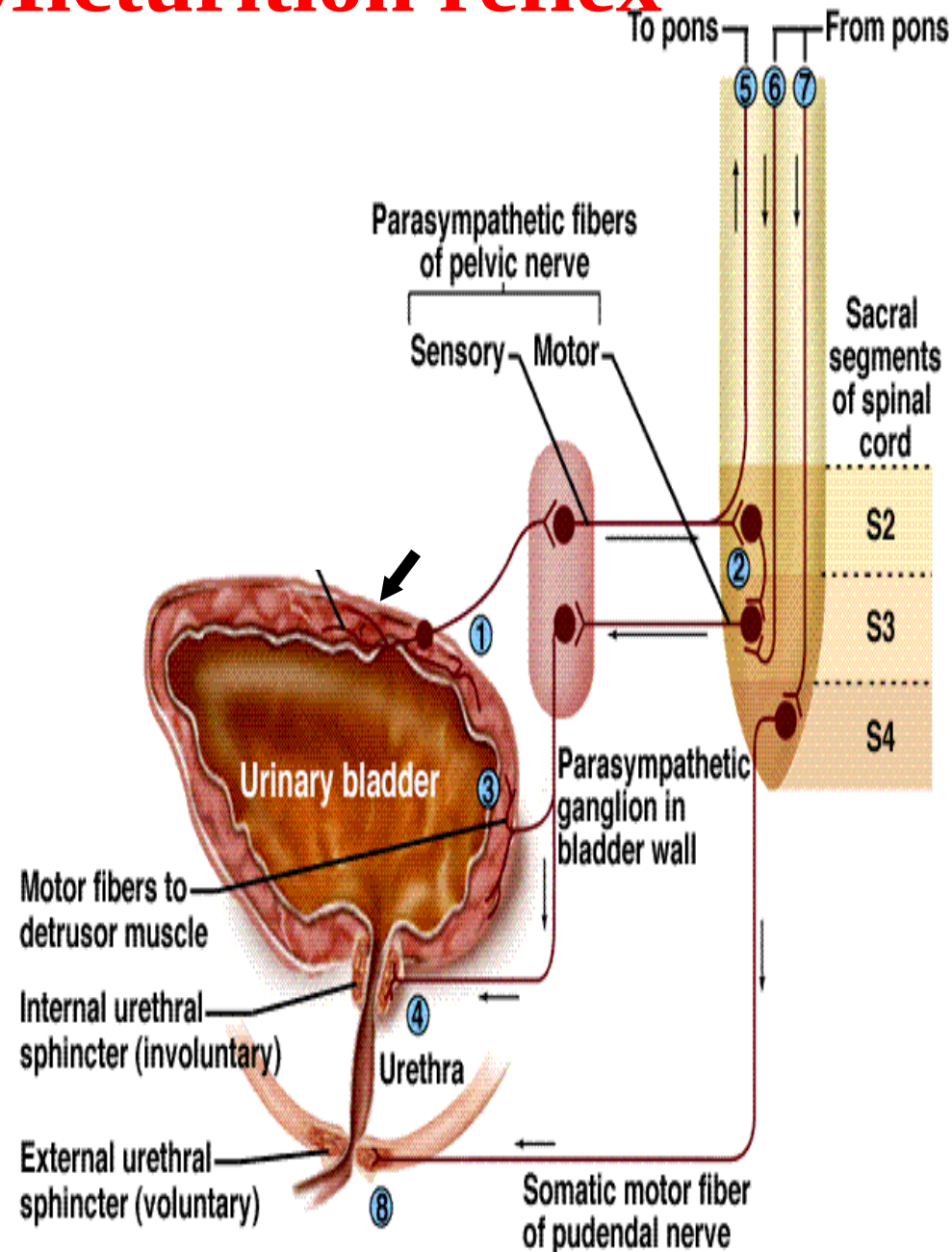


Urine stimulates stretch receptors → AP through sensory neurons to → sacral spinal cord → reflex back through PSN to bladder → bladder is contracted → urine released through urethra

➤ Stretch receptors also send impulse to higher centers (Pons, HT and cerebral cortex)

1. Motor impulse from higher centers promote readiness to urinate
2. Identify places for urination
3. Relax external urethral sphincter

Micturition reflex



Urethra

- Muscular tube that:
 - Drains urine from the bladder
 - Moves urine out of the body
- Sphincters keep the urethra closed when urine is not being passed
 - **Internal sphincter** – involuntary sphincter at the bladder-urethra junction
 - **External sphincter** – voluntary sphincter surrounding the urethra as it passes through the urogenital diaphragm

Concept of clearance

- **Clearance**: is a measure of the rate at which the kidneys remove, i.e., “clear”, a substance from the plasma.

- Clearance (Cx) equation is:

$$C_x \text{ (ml/min)} = \frac{U_x * V_x}{P_x}$$

- U_x= The conc. of a substance (X) in urine (mg/ml)
 - V_x = Urine volume/min in (ml/min)
 - P_x = The conc. of the substance (X) in plasma (mg/ml)
- Clearance of inulin and Creatinine is almost equal to GFR

- **Clearance** has a very important implication clinically, b/s it indicates the **efficiency of the kidneys to clear the plasma of a given substance.**
- By using clearance method, one can calculate:
 - **GFR (glomerular filtration rate)**
 - **RBF (Renal blood flow) etc.**
 - It can also be used to know how different substances are **reabsorbed or secreted**

Inulin clearance

- Inulin clearance can be **used to estimate GFR**.
- This is because, Inulin is **neither reabsorbed nor secreted** by the nephron and **cannot be metabolized**.

Therefore

- The **rate of inulin filtration** equals the rate of inulin **excretion** in urine.
 - so the **volume of plasma cleared per minute** **equals GFR**.

By using inulin clearance

- A clearance value of **less than 125 ml/min** indicates **reabsorption**
- And **greater 125ml/min** indicates **secretion**
- **Inulin usually is not used clinically because it must be intravenously infused (toxicity).**

Creatinine and Glucose clearance

- Creatinine clearance:

- Cr. is a **product of muscle metabolism**
- can be used instead of **inulin**
- overestimates GFR by about **10%** b/c there is **some secretion into the proximal tubule.**

- Glucose Clearance

- Glucose like inulin is freely filtered, but all the glucose filtered is reabsorbed.
- No plasma ends up losing glucose.
 - **Thus the clearance of glucose is zero (0).**

Example:-

- If the *plasma* [urea] is **0.25 mg/mL**, the *urinary* [urea] is **10 mg/mL**, and the urine flow rate is **1 mL/min**, calculate the clearance rate of urea.
- $C_{urea} = U_{urea} \times V_{urine} / P_{urea}$
 $= \frac{10 \text{ mg/mL} \times 1 \text{ mL/min}}{0.25 \text{ mg/ml}}$
- *Clearance* = **40 mL/min**

- **A clearance of 40 mL/min means that 40 mL of plasma is required to supply the urea lost in urine every minute.** b/s
- **Clearance is a volume of plasma “cleaned” per unit time (mL/min), not the amount of solute excreted in urine per unit time (mg/min).**

Acid-base balance

- The body regulates acid base imbalances by 3-ways that include:

- Chemical buffers

- The lung (CO_2)

- Kidney (HCO_3^-)

Chemical buffers

- Chemical buffers regulate with in seconds and is the 1st line of defense.
- Its capacity is limited by the amount of body buffers
- 4-types of chemical buffers:
 1. Bicarbonate buffer (ECF)
 2. Protein buffers (ICF)
 - 3 .Hemoglobin (ICF)
 4. Phosphate (ICF)

The bicarbonate buffer is the most important.



The ratio $\frac{\text{HCO}_3}{\text{CO}_2} = \frac{20}{1}$

Metabolic acidosis

Causes:-

- **Starvation**: mostly fats burn instead of carbohydrates resulting **ketoacidosis**
- **Hypoxia**: decreased O_2 , thus glycolysis dominates and **lactic acid** is generated
- **Severe diarrhea**: **large amounts of HCO_3^- is lost, acid dominates.**

Compensation - Is usually done **through respiratory means.**

– **hyperventilation** - causes washout of CO_2 and decreases H^+ indirectly from body fluids.

$H^+ + HCO_3^- = H_2CO_3 = CO_2 + H_2O$ (the reaction goes to the right and CO_2 is eliminated)

Respiratory acidosis

Cause - hypoventilation

- Failure of lungs to expire CO₂ produced by metabolism
- Hypoventilation caused by:
 - a. **Respiratory depression** (e.g., drugs, sedatives)
 - b. **Pulmonary diseases** – like emphysema, asthma etc.

Compensation - By the kidneys:

- It increases the reabsorption of HCO₃.
- HCO₃ raises the pH back to normal.

Metabolic Alkalosis

Cause:

- a. **Prolonged vomiting**: HCl acid from the stomach is lost and this effect elevates blood HCO_3^- level.
- b. **Hyper-aldosteronism**: increase HCO_3^- reabsorption

Compensation

- The respiratory system inhibits respiration (causes **lower breathing rate**)
- **Hypoventilation** causes retention of CO_2 and causes the pH to come to normal

Respiratory alkalosis

Causes:

- Excessive removal of CO₂ from the body through **hyperventilation**
- This causes a drop in PCO₂ level

• Compensation

- Kidney results in a **loss of HCO₃ in urine** until pH becomes normal

The END